

On-chip heralded single photon sources EP

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ABSTRACT

Time correlated photon pairs are used to produce heralded single photon states for quantum integrated circuits. These states are generated by photon sources, which are called heralded single photon sources. They are based on the detection of one photon which heralds the presence of the other. In this way, pure single photon states can be probabilistically generated and, subsequently, manipulated in complex photonic circuits. Heralded single photon sources are the topic of this review. The authors detail the main parameters and the experiments involved in their characterization, with a focus on their use in integrated photonic circuits. Different geometries and technological platforms are compared, and an assessment of their performances is performed. The final result is that nearly perfect single photon sources are possible and feasible in an integrated platform. A few open issues are also underlined.

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TABLE OF CONTENTS

I. INTRODUCTION	1
II. SINGLE PHOTONS	2
A. Basics of single photon sources	2
B. Heralded parametric sources	5
III. HERALDED SINGLE PHOTON PARAMETERS AND CHARACTERIZATION	6
A. Purity	6
B. Joint spectral intensity	9
C. Antibunching dip	10
D. Heralding efficiency	11
E. Brightness	12
F. Coincidence to accidental ratio	12
G. Indistinguishability	13
1. Joint spectral intensity overlap	13
2. Reversed Hong–Ou–Mandel	14
3. Quantum state tomography	15
H. Quantum interference	15
IV. STATE-OF-THE-ART	15
A. Optical waveguides	16
B. Optical microresonators	18
C. Periodically poled waveguides	21
D. Multiplexed heralded sources	21
V. CONCLUSION	24

I. INTRODUCTION

Many technological innovations in the next decade will likely be based on quantum physics. Indeed, quantum science and technology

could experience a revolution, in terms of new discoveries, new phenomena, new applications, and improved performances.¹ Communication,² computation,³ sensing,⁴ and metrology⁵ are only a few of the many areas that benefit from the quantum paradigm. In order to lay the foundation of such quantum architecture, different physical systems can be used, e.g., trapped ions,⁶ cold gases,⁷ spins,⁸ solid state Josephson junctions,⁹ and single photons.¹⁰ Very recently, quantum supremacy has been claimed in a Josephson junction based quantum computer.¹¹ Still, this quantum computer requires cryogenic temperatures, high vacuum, and bulky and complicated setups. On the other hand, commercial devices which take advantage of quantum components are appearing in the market.¹² With respect to a widespread diffusion of quantum technologies, the photonic platform—where quantum devices are fabricated by photonic integrated circuits (PIC)—is more promising.¹³ Being weakly coupled to the environment, photons are affected by limited decoherence, enabling room temperature operation. Photonics guarantees also high phase stability between the propagating optical modes involved in quantum operations, which is essential in large scale quantum circuits.¹⁴ Moreover, photonic qubits benefit from the already mature fiber based communication network, with the potential of global quantum communication applications.¹⁵ Both distributed quantum networks and quantum computing devices require a large number of components.¹⁴ Therefore, integrated photonics is the ideal platform to develop such quantum architectures.¹⁶ Among the other integration platforms, owing to its dense device fabrication and compatibility with the mature CMOS (complementary metal-oxide semiconductor)

technology, silicon photonics¹⁷ appears to be the best candidate to scale-up quantum optical technologies.^{18,19} To unveil quantum optical phenomena at the chip scale, such as teleportation, quantum discord, entanglement, or quantum superposition, single photons have to be generated and manipulated in the PIC. Quantum manipulation benefits from the mature linear components already developed for the classical (i.e., based on the Maxwell equations) integrated photonic circuits.¹⁷ Phase shifters, directional couplers, Mach-Zehnder interferometers (MZI), microring resonators, and AWG (array waveguide grating) (de)multiplexers are easily designed and can be used as the building blocks of quantum photonic operations to encode, manipulate, and read the results of quantum information processing on the photons.^{14,20} On the other hand, the generation of single photons is still an open challenge.

The ideal single photon source emits deterministically exactly one and only one photon at a time with 100% generation probability.²¹ This is the concept of the “photon gun,”²² where pressing a button causes the instantaneous emission of a single photon with unit probability. There are two main categories of single photon sources: deterministic single photon emitters and probabilistic heralded single photon sources. Single photon emitters are the closest example of a photon gun. Examples in this area are color centers in diamonds,²³ defects in crystals,²⁴ and quantum dots.²⁵ All these systems are based on atomic-like emitters which are described by a two-level energy diagram: a single photon is emitted when the high energy level is populated. The photon emission occurs via a single optical transition, which prevents the generation of more than one photon in the process. Semiconductor quantum dots are particularly appealing as single photon sources because they can be electrically operated. They have provided breakthrough results in quantum experiments^{26,27} and for single photon quality.^{28,29} Despite the high performance of quantum dot sources, their integration in a PIC is still an open issue due to high coupling losses, elaborated fabrication processes, and the need for cryogenic temperatures.^{30,31} Moreover, size fluctuations and emission wavelength dispersion due to the limits in their fabrication impact negatively on the production of identical single photon sources.³⁰ This is detrimental when implementing quantum protocols, where many indistinguishable single photons are mandatory.³² The state-of-the-art results indicate indistinguishability among the generated sources of up to 92%.³³ The limited reproducibility and scalability of single-photon emitters, together with the typical needs to work at cryogenic temperatures,^{30,34,35} pushed the search for alternative single photon sources.

An alternative to single photon emitters is the heralding of single photon states. This is based on the temporal correlation between photon pairs generated by means of spontaneous parametric nonlinear processes, such as spontaneous parametric down conversion (SPDC) or spontaneous four wave mixing (SFWM).³⁶ Photons generated through parametric spontaneous processes are emitted at the same time. Then, one of the two photons is detected to herald the presence of the other. The detection of the herald photon confirms that the pair generation has occurred, and that the remaining twin photon, i.e., the heralded photon, is ready to be used as a single photon in the quantum device. However, heralded sources are intrinsically probabilistic, being based on spontaneous nonlinear parametric processes. Thus, these sources lack on-demand generation possibility. Nevertheless, the probabilistic nature of heralded single photon sources can be mitigated by means of multiplexing, where single photons heralded from different

spatial, temporal, or frequency modes are combined together to produce a beam of near-deterministic single photons.^{37–40}

All-optical quantum architectures are based on large scale circuits, with many identical single photon emitters coupled to plenty of photonic components. These requirements, at present, are best met by fully integrated heralded single photons, routed by a state-of-the-art silicon-on-insulator (SOI) or glass based optical circuitry.^{19,41,42} The potential of the integrated quantum photonic platform is repeatedly confirmed by continuous experimental validations.^{41,43–45} In spite of this, the generation rate and performances of heralded single photon sources are limiting further scaling-up and demonstration of quantum photonic architectures and protocols.^{32,46} This motivates the research for better on-chip heralded single photon sources.^{18,47–50} It is therefore the right time to make the point on the development and characterization of heralded single photon sources. In this review, after the definition and the description of a single photon source (Sec. II), we discuss the main parameters to assess its performance as well as the experimental strategies to measure them (Sec. III). Some of these parameters are not uniquely defined in the literature, and different metrics are reported; this makes it difficult to compare different devices or to investigate the best strategy for performance characterization. Here, we aim to clarify these terms, providing a list of parameters for heralded single photon source characterization. An overview of the current state-of-the-art is presented with the description of the various platforms for PIC (Sec. IV). Ongoing challenges are also explored, focusing on multiplexing solutions for bright and near-deterministic single photon emission (Sec. IV D). Section V concludes the work by summarizing the main points and assessing the various approaches.

II. SINGLE PHOTONS

A. Basics of single photon sources

Let us describe a pulse of light in the quantum optics framework where the light pulse is decomposed into photon modes (polarization, frequency, or spatial modes).⁵¹ If we assume single modes, the light state vector is given by a linear combination of Fock states (or number states) $|n\rangle$. Therefore, the states generated by a single photon emitter ($|\psi\rangle_{\text{single}}$) or by a parametric photon pair source ($|\psi\rangle_{\text{pair}}$) are, respectively, written as

$$|\psi\rangle_{\text{single}} = \sum_{n=0}^{\infty} \zeta_n |n\rangle, \quad (1a)$$

$$|\psi\rangle_{\text{pair}} = \sum_{n=0}^{\infty} \nu_n |n\rangle_a |n\rangle_b. \quad (1b)$$

In $|\psi\rangle_{\text{single}}$, n photons occupy the states of the single-mode beam with probability $|\zeta_n|^2$, i.e., the single photon emitter generates n photons with probability $P(n) = |\zeta_n|^2$. Thus, $P(n)$ describes also the photon number distribution generated by the source. In $|\psi\rangle_{\text{pair}}$, two single-mode beams, a and b , are involved. If losses and noise are neglected, the number of photons in a and b is the same due to the nature of the pair generation process and the probability to find n photons in a and n in b is $|\nu_n|^2$, with $P(n) = |\nu_n|^2$. The ideal single photon source produces, deterministically, one and only one photon per mode ($P(1) = 1$). Thus, its vector state is $|\psi\rangle = |1\rangle$. This means that it does not involve $|n \neq 1\rangle$ number states (also the vacuum state $|0\rangle$ is empty). Analogously, in the case of the ideal deterministic photon pair source, the vector state is $|\psi\rangle_{\text{pair}} = |1\rangle_a |1\rangle_b$ ($P(1) = 1$ here as well). It

is clear that the main peculiarity of an ideal photon pair source is the simultaneous generation of photons in a and b . This temporal correlation is exploited in the heralding mechanism. Being the timing information of a encoded in b , the photon measured in b heralds the existence and the temporal position of that in a and vice versa. Note that an ideal parametric source produces a thermal distribution of photon number states and, therefore, would not produce the pure state $|1\rangle_a|1\rangle_b$.

Actual single photon and photon pair sources show nonunity emission efficiency and non-negligible multiphoton emission. As a result, $P(0)$ and $P(n > 1)$ start to play a role. In Fig. 1, qualitative plots of the emission statistics, i.e., of the photon number distribution $P(n)$, are reported for (a) the ideal single photon emitter, (b) the two-level atom-like emitter, (c) the parametric photon pair source, and (d) the heralded single photon source. The ideal source exhibits perfect single photon emission, with $P(1) = 1$, meaning the absence of multiphoton emission and no failed attempts. The two-level atom-like emitter resembles the ideal one, with $P(1)$ dominating over $n=0$ and $n > 1$ probabilities. The parametric photon pair source, on the contrary, shows significant vacuum state occupation and multiphoton states. Figure 1(d) shows the effect of the heralding process. $P(n)$ is now the results of the conditional probability to have photons in a mode conditioned to the presence of photons in the other mode. By heralding, in fact, we look at the emitted photons (heralded beam) only when we already know that the photons have been emitted. In this way, the probability to measure a vacuum emission [$P(0)$] is dramatically reduced, yielding an almost single photon statistics.

From this reasoning, two rules to approximate the single photon generation by a heralded single photon source can be derived:

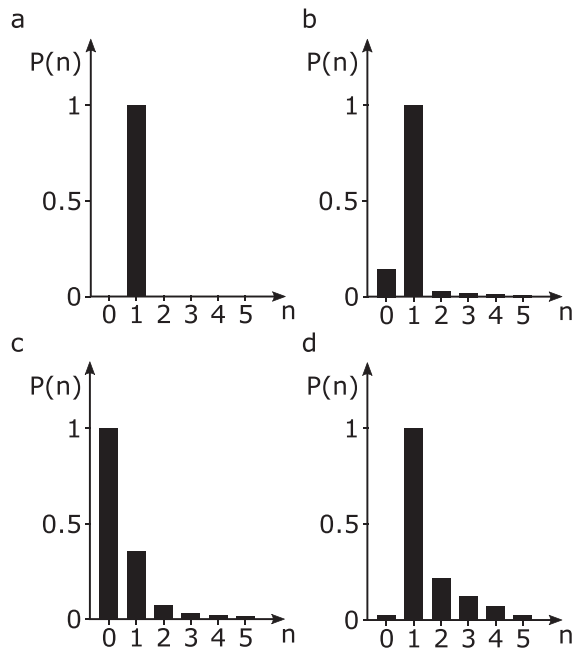


FIG. 1. Normalized photon number distribution for (a) an ideal single photon emitter, (b) a two-level atom-like emitter, (c) a single beam of the photon pair source, and (d) the heralded beam of the photon pair source.

1. $P(0) \rightarrow 0$, meaning that vacuum emissions are removed;
2. $P(n > 1) \rightarrow 0$, meaning that one and only one photon is measured at a time.

It follows that it is mandatory to resolve the whole photon number distribution to characterize the single photon emission character of a source. This requires photon number resolving (PNR) detectors, that are not the standard in optics experiments.²¹ The most widely adopted single photon detectors (SPDs) in quantum optics experiments are click/no-click detectors.⁵² Such devices detect the arrival of photons, but do not resolve their number. Therefore, to measure the photon statistics, one has to use a Hanbury–Brown and Twiss (HBT) interferometer (Fig. 2).⁵³

Briefly, in a HBT interferometer, two single photon detectors (D_3 and D_4) are placed in the transmission (4) and reflection (3) ports of a beam splitter (BS), with the input light in port 1 and vacuum in port 2. The detectors which measure the arrival of photons are connected to a counting electronic device, which stores coincidence events and their timing. By performing correlation measurements between the clicks of the two detectors, the degree of second order coherence $g^{(2)}(\Delta t)$ is estimated, with Δt being the time delay between the clicks of the two detectors. $g^{(2)}$ yields the statistics of the light source (Fig. 3). It is easy to demonstrate that $g^{(2)}$ is equal to²¹

$$g^{(2)}(\Delta t) = \frac{\langle \hat{a}^\dagger(t) \hat{a}^\dagger(t + \Delta t) \hat{a}(t + \Delta t) \hat{a}(t) \rangle}{\langle \hat{a}^\dagger(t) \hat{a}(t) \rangle \langle \hat{a}^\dagger(t + \Delta t) \hat{a}(t + \Delta t) \rangle}, \quad (2)$$

where the angled brackets denote an ensemble average and $\hat{a}^\dagger(t)$ and $\hat{a}(t)$ are, respectively, the photon creation and destruction operators at time t . Notice that $\langle \hat{a}^\dagger(t + \Delta t) \hat{a}(t + \Delta t) \rangle = \langle \hat{a}^\dagger(t) \hat{a}(t) \rangle$, because the ensemble average is time invariant. Considering the zero-delay case $\Delta t = 0$, Eq. (2) can be rewritten in terms of the photon number operator $\hat{n} = \hat{a}^\dagger \hat{a}$

$$g^{(2)}(0) = \frac{\langle \hat{n}(t) (\hat{n}(t) - 1) \rangle}{\langle \hat{n}(t) \rangle^2}. \quad (3)$$

$\hat{n}$ is such that $\hat{n}(t)|n\rangle = n(t)|n\rangle$. From Eq. (3), we can express $g^{(2)}(0)$ as a function of the mean number of photons in the mode $\bar{n} = \langle \hat{n} \rangle$

$$g^{(2)}(0) = \frac{\langle \hat{n}^2 \rangle - \langle \hat{n} \rangle}{\langle \hat{n} \rangle^2} = 1 + \frac{\Delta n^2 - \bar{n}}{\bar{n}^2}, \quad (4)$$

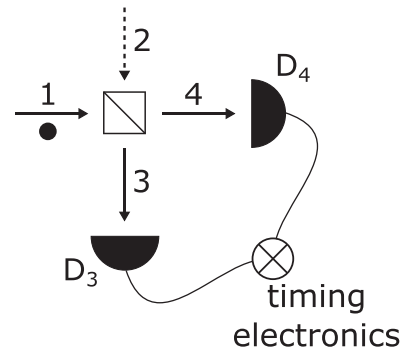


FIG. 2. Basic Hanbury–Brown and Twiss interferometer setup.

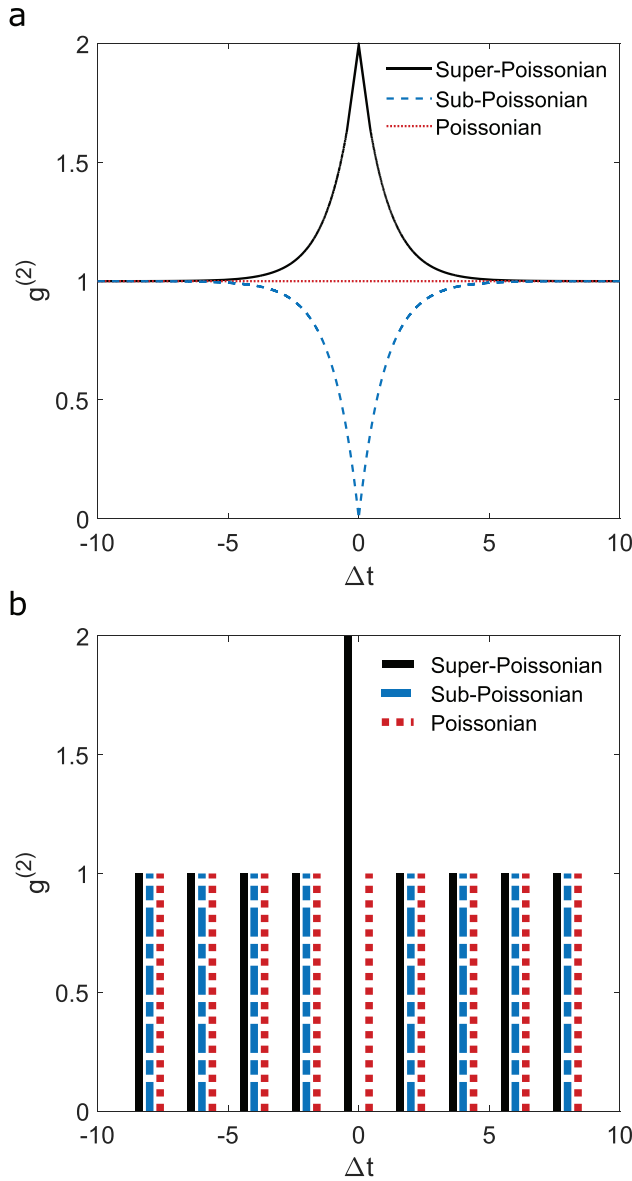


FIG. 3. Degree of second-order coherence $g^{(2)}(\Delta t)$ for various light sources: super-Poissonian (black), sub-Poissonian (blue), and Poissonian (red). (a) CW light and (b) pulsed light.

where $\Delta n^2 = \langle \hat{n} - \langle \hat{n} \rangle^2$ is the photon number fluctuations. Therefore, Eq. (3) or Eq. (4) relates the number of photons in the single mode beam with the experimental measurement of the second order coherence. Because of this, $g^{(2)}(0)$ is the most relevant parameter in defining the source statistics.

Coherent light has a photon statistics which is Poissonian, i.e., photon arrival times at detectors 3 and 4 shown in Fig. 2 are distributed randomly and the probability of detecting a photon is independent of when any other photon arrives. For Poissonian statistics, the variance is equal to the mean. In this case, Eq. (4) shows that $g^{(2)}(0) = 1$. Photons are completely uncorrelated. If $g^{(2)}(0) > 1$, then

the fluctuations in the photon number are larger than the average photon number and the photon statistics becomes super-Poissonian. In this case, the large fluctuations cause the photons to reach the detectors in bunches, an effect which is peculiar of thermal light. For Poissonian and super-Poissonian light, the $g^{(2)}$ is flat or peaked at $\Delta t = 0$, respectively, as shown in Fig. 3. The peak in the super-Poissonian $g^{(2)}$ is characteristic of the bunching behavior of the source, with photons emitted more probably closely spaced in time. The opposite of bunching is antibunching, where fewer photons are emitted close together than further apart. In this case, since one photon cannot be split, one single detector clicks at a time in a HBT experiment. The beam carries one photon at a time, with $g^{(2)}(0) \simeq 0$. In fact, considering a Fock state with exactly n photons, and neglecting the temporal dependence of $\hat{n}$, Eq. (3) becomes

$$g^{(2)}(0) = \frac{\langle n|\hat{n}(\hat{n}-1)|n\rangle}{\langle n|\hat{n}|n\rangle^2} = 1 - \frac{1}{n}, \quad (5)$$

thus, it is easy to guess the experimental outcome of $g^{(2)}(0)$ when a n -photon beam is analyzed. For example, with single photon light, $n=1$, $g^{(2)}(0)=0$ as expected. Whenever $g^{(2)}(0) < 1$, the source emits antibunched light with a sub-Poissonian statistics. This is characteristic of a nonclassical nature of light. Antibunching is revealed by a dip in the $g^{(2)}(\Delta t)$, as shown in Fig. 3. However, the evidence of antibunching is not enough to claim single photon emission. In fact, with $n=2$, $g^{(2)}(0) = 0.5$. This means that, to assert the single photon character of a light source, the measured zero-delay $g^{(2)}(0)$ should be lower than 0.5.⁴⁸

Therefore, the $g^{(2)}(\Delta t)$ is the most important metrics to assess the single photon character of a source. The HBT experiment is particularly suited for its estimation, thanks to these properties:²¹

- unbalanced transmission and reflection coefficients of the beam splitter do not affect the $g^{(2)}$ measurement;
- the two detectors do not need 100% efficiency or balanced efficiencies;
- click detectors are suitable for the HBT experiment without losing accuracy if $P(1) \gg P(2) \gg P(n > 2)$;
- losses do not modify the $g^{(2)}$ measurement.

A final remark on the experimental measurement of $g^{(2)}(\Delta t)$ is about the pulsed or continuous wave (CW) character of the source. Equation (2) is written in the CW case. The pulsed case is analogous, with the continuous time variable t replaced by an integer variable denoting the pulse number, such that Eq. (2) becomes

$$g^{(2)}[k] = \frac{\langle \hat{a}^\dagger[m] \hat{a}^\dagger[m+k] \hat{a}[m+k] \hat{a}[m] \rangle}{\langle \hat{a}^\dagger[m] \hat{a}[m] \rangle \langle \hat{a}^\dagger[m+k] \hat{a}[m+k] \rangle}, \quad (6)$$

with m and k integer numbers indicating the number of the pulse and the pulse delay, respectively. Specifically, $t \equiv mT$, with T being the period between pulses, and $\Delta t = kT$. Figure 3(b) shows the expected experimental $g^{(2)}[k]$ for the various sources discussed so far. Note that for a single photon source $g^{(2)}[0] = 0$. From now on, we will use a continuous t variable independent of the pulsed or CW character of the source.

$g^{(2)}$ can be directly related to the photon number distribution via the density matrix operator $\hat{\rho}$. Indeed, the ensemble average of $\hat{n}$ can be written as

$$\langle \hat{n} \rangle = \text{Tr}\{\hat{\rho}\hat{n}\} = \mu, \quad (7)$$

where μ is the average number of photons per pulse. Using the photon number states as basis states, the photon number probabilities $P(n)$ are encoded in the diagonal elements of the density matrix,⁵⁴ such that

$$P(n) = \text{Tr}\{\hat{\rho}|n\rangle\langle n|\} = \langle n|\hat{\rho}|n\rangle, \quad (8)$$

and the density matrix, with such a basis, is given as

$$\hat{\rho} = \sum_{n=0}^{\infty} P(n)|n\rangle\langle n|. \quad (9)$$

Therefore, Eq. (3), in terms of photon number probabilities, reads

$$g^{(2)}(0) = \frac{\text{Tr}\{\hat{\rho}\hat{n}(\hat{n}-1)\}}{(\text{Tr}\{\hat{\rho}\hat{n}\})^2} = \frac{\sum_{n=0}^{\infty} n(n-1)P(n)}{\left[\sum_{n=0}^{\infty} nP(n)\right]^2}. \quad (10)$$

If the source is characterized by $P(1) \gg P(2) \gg P(n > 2)$, which is often the case when dealing with low-efficiency sources, the following expression can be derived:²¹

$$g^{(2)}(0) \simeq \frac{2P(2)}{\mu^2} \simeq \frac{2P(2)}{P(1)^2}, \quad (11)$$

where $\mu \simeq P(1)$ comes from the assumption $P(1) \gg P(2)$. Equation (11) states that as soon as the source of single photons is characterized by $P(1) > \sqrt{2P(2)}$, a $g^{(2)}(0) < 1$ occurs, approaching zero as $P(1)^2/P(2)$ increases.

B. Heralded parametric sources

In a PIC, a photon pair emitter can be used as a single photon source, thanks to the heralding mechanism. In this case, photon pairs are generated by optical nonlinear processes in a waveguide.³⁶ To model the nonlinear response of a material, the polarization vector $\mathbf{P}$ is usually expanded in a Taylor series,⁵⁵

$$\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_3 + \dots \quad (12)$$

$\mathbf{P}_1$ describes the linear response. $\mathbf{P}_m$, with $m = 2, 3, \dots$, are the nonlinear terms. The series is usually truncated at the third term, with $\mathbf{P}_{m>3}$ values typically negligible for photonics. The nonlinear terms in the Taylor expansion of the polarization vector are given by

$$\mathbf{P}_2 = \epsilon_0 \chi^{(2)} \cdot \mathbf{E}^2, \quad (13a)$$

$$\mathbf{P}_3 = \epsilon_0 \chi^{(3)} \cdot \mathbf{E}^3, \quad (13b)$$

with ϵ_0 being the vacuum permittivity, $\chi^{(2)}$ and $\chi^{(3)}$ being the second order and the third order nonlinear susceptibilities, and $\mathbf{E}$ being the electric field. Centrosymmetric materials display only third order nonlinear interactions, while noncentrosymmetric materials may also show $\chi^{(2)}$ -processes.⁵⁶ Among the most known $\chi^{(3)}$ materials, we find silicon, silicon nitride, and silica. Lithium niobate, barium titanate (BTO), gallium arsenide, and aluminium nitride are some $\chi^{(2)}$ materials best known for their integration possibilities. Recently, a sizable $\chi^{(2)}$ was induced in silicon by strain⁵⁷ or by a static electric field.⁵⁸ The parametric nonlinear

processes involved in the generation of photon pairs are SPDC ($\chi^{(2)}$) and SFWM ($\chi^{(3)}$). SPDC and SFWM have in common the general mechanism, in which a strong pump field is injected in the waveguide, and a pair of photons called the signal and idler is generated. The signal is typically the shorter wavelength photon and the idler is the longer wavelength photon. However, this standard is not always respected. Both the SPDC and SFWM processes are governed by energy conservation and momentum conservation. The former is written as

$$\omega_p = \omega_s + \omega_i \quad (\text{SPDC}), \quad (14a)$$

$$\omega_{p1} + \omega_{p2} = \omega_s + \omega_i \quad (\text{SFWM}), \quad (14b)$$

with ω_j ($j = p, p1, p2, s, i$) being the frequencies of the pumps and signal and idler photons. The momentum conservation, also known as the phase matching condition, is given by

$$k_p = k_s + k_i \quad (\text{SPDC}), \quad (15a)$$

$$k_{p1} + k_{p2} = k_s + k_i \quad (\text{SFWM}), \quad (15b)$$

with k_j ($j = p, p1, p2, s, i$) being the wavevectors of the pumps and signal and idler photons. Note that in a waveguide, these wavevectors are the propagation constants β_j of the associated optical modes. $\beta_j = n_{\text{eff}}(\omega_j)\omega_j$, where $n_{\text{eff}}(\omega_j)$ is the effective index of the j th mode. Energy conservation governs the spectral correlations between the involved photons, while phase matching determines the spectral bandwidth and position where efficient pair generation occurs.⁵⁵ While energy conservation is naturally fulfilled in the process, phase matching has to be engineered in order to give efficient and controllable generation. Here, the main problems are associated with the dispersion of the material refractive index. The integrated platform offers a great opportunity in tailoring such a parameter, leveraging on the dependence of the photon propagation constants on the waveguide geometry. As a result, the phase matching condition can be easily tailored, exploiting the chromatic dispersion of the waveguide modes.⁵⁹

In more detail, in order to achieve an efficient parametric generation, the phase mismatch $\Delta\beta$ between the different propagating photons in the waveguide must be near to zero. For brevity, let us concentrate on the SFWM of degenerate pump photons ($\omega_{p1} = \omega_{p2}$). Neglecting other nonlinear effects (e.g., cross phase modulation, self-phase modulation, etc.),³⁶

$$\Delta\beta = 2\omega_p n_{\text{eff}}(\omega_p) - [\omega_s n_{\text{eff}}(\omega_s) + \omega_i n_{\text{eff}}(\omega_i)], \quad (16)$$

n_{eff} and thus β_j can be engineered acting on the waveguide geometric dimensions. Let us use the notation $\beta_n = \partial^n \beta(\omega)/\partial \omega^n$ for the n th order term in the expansion of $\beta(\omega)$ as a Taylor series around ω . Then, we can also expand the phase mismatch parameter $\Delta\beta$ around the pump frequency. Since the odd power terms in $\Delta\omega = \omega_p - \omega_s = -\omega_p + \omega_i$ are canceled out for symmetry reasons, $\Delta\beta$ at the second order in $\Delta\omega$ is given by

$$\Delta\beta \simeq \beta_2(\omega_p)(\Delta\omega)^2, \quad (17)$$

where β_2 , the second order derivative of the propagation constant, is the group velocity dispersion (GVD) parameter of the waveguide. Therefore, phase matching can be assured if $\beta_2 \sim 0$ at the pump frequency. Another possibility to achieve phase matching is suggested by expanding $\Delta\beta$ to the following order in $\Delta\omega$:

$$\Delta\beta \simeq \beta_2(\omega_p)(\Delta\omega)^2 + \frac{\beta_4(\omega_p)}{12}(\Delta\omega)^4. \quad (18)$$

The waveguide dimensions can be engineered to achieve a GVD parameter β_2 and a fourth order dispersion parameter β_4 with an opposite sign and with the proper magnitude to have $\Delta\beta \sim 0$.⁶⁰

Another opportunity is offered by the use of multimodal waveguides.⁶¹ In this approach, the photons propagate on different modes with different dispersion relationships. In this way, by expanding the propagation constants of the three different modes around the pump frequency, it is possible to write⁶²

$$\Delta\beta \simeq \Delta\beta_0 + \Delta\omega(\beta_{1,i} - \beta_{1,s}), \quad (19)$$

where $\Delta\beta_0 = 2\beta_p(\omega_p) - \beta_i(\omega_p) - \beta_s(\omega_p)$ is the phase mismatch at the pump frequency and $\beta_{1,j}$ is related to the group velocity of the j mode. The basic idea of the inter-modal phase matching technique is then to balance the $\Delta\beta_0$ parameter by means of the difference between the signal and idler group velocities. As a result, discrete band photon pair generation can be achieved, where the signal and idler photon wavelengths can be very different.⁶³ Note that Eq. (19) neglects high order terms and a better approach to design waveguides with $\Delta\beta \simeq 0$ is to numerically solve Eq. (16).

Finally, alternative techniques are based on quasi-phase matching (QPM) conditions.⁶⁴ In this case, one exploits a periodic modulation in the waveguide parameters (poling) so that

$$\Delta\beta = \Delta\beta_0 + \frac{2\pi}{\Lambda}, \quad (20)$$

where Λ is the periodicity of the poling. Λ is determined by the condition that $\Delta\beta = 0$.

In principle, the photon pairs generated in parametric processes like SFWM and SPDC are correlated in time and energy. The photon pair state $|\text{II}\rangle$ generated in SFWM (or SPDC) is the tensor product of independent two-mode squeezers^{65,66}

$$|\text{II}\rangle = \hat{S}_{si}^m(\xi_m)|0\rangle_s|0\rangle_i, \quad (21)$$

where $\hat{S}_{si}^m$ is the two-mode squeezing operator (in short squeezer) and ξ_m is the squeezing parameter. The squeezing parameter depends on the pump intensity and on the nonlinearity strength. Each squeezer acts on the signal (s) and idler (i) modes, operating on the same spectral and temporal modes. In the limit of single mode emission, which is the ideal condition for heralded single photons, Eq. (21) takes the form

$$|\text{II}\rangle = \hat{S}_{s,i}(\xi)|0\rangle_s|0\rangle_i = \sum_n \nu_n |n\rangle_s |n\rangle_i, \quad (22)$$

that is analogous to Eq. (1b), the ideal photon pair emitter state. From Eq. (22), two fundamental points are evident:

1. the n pairs are generated in the same temporal mode;
2. the idler and signal photon number probabilities are the same, $P_s(n) = P_i(n) = P(n) = |\nu_n|^2$.

These conclusions lie at the basis of the heralding mechanism, where by detecting n photons in the signal mode, n photons are heralded in the idler mode, or vice versa. The conditional detection of the heralded photons increases the mean photon number μ in Eq. (7),

because $P(0)$ goes to zero. This can be understood considering the conditional probability of heralding n photons, which can be written as

$$P^{(h)}(n) = \frac{P(n \cap h)}{P(h)}, \quad (23)$$

with $P(n \cap h)$ being the joint probability of detecting n heralded photons and h heralding photons and $P(h)$ being the probability of detecting h heralding photons.⁶⁷ With $h \neq 0$, $P^{(h)}(0) = 0$, meaning that, given a heralding detection, it is not possible to measure a vacuum heralded state. This is a consequence of the correlation between the paired photons. As a result, all the conditional probabilities $P^{(h)}(n > 0)$ increase with respect to the unconditional ones. In the realistic case where $P^{(h)}(1) \gg P^{(h)}(2)$, $\mu \simeq P^{(h)}(1)$. Heralding changes the individual photon number probabilities, but their ratio remains constant. Therefore, according to Eq. (11), the conditional $g^{(2)}(0) \ll 1$, because $P^{(h)}(2)/P^{(h)}(1)$ is not changed, but $P^{(h)}(2)/P^{(h)}(1)^2$ is effectively decreased to zero. Here resides the essence of heralding: time correlated photons that lead to single photon emission statistics. From now on, we use the photon number probabilities $P(n)$ without distinguishing between conditional or not conditional probabilities, since they differ only for a normalization factor.

III. HERALDED SINGLE PHOTON PARAMETERS AND CHARACTERIZATION

The heralding performance is quantified through several metrics (see Table I).

A. Purity

Quantum information usually requires a high visibility quantum interference between single photons.⁶⁸ To achieve this, the interfering single photons must be individually in a pure state. Heralded single photon sources generate pure photons when the herald photon detection does not perturb the heralded photon. Usually this is not the case, since nonlinear parametric processes generate photon pairs in an entangled state. Such a high degree of correlation between the twin photons causes the heralded photon to be projected into a mixed state when the herald one is detected.⁵⁴ Purity measures the amount of mixedness of a state, i.e., how far it is from a pure state. Perfectly pure heralded single photons have purity $P = 1$, while maximally entangled photons have $P = 0$. Purity can therefore be considered as a measure of the subsisting correlations between the twin photons.

Let us consider the states generated in a parametric process, e.g., degenerate SFWM. Assume that the SFWM occurs in a microring resonator with many different resonances, and let us count them from the one used to pump the system (i.e., ..., ω_{-2} , ω_{-1} , ω_p , ω_1 , ω_2 , ...).⁴⁸ The two-photon state produced via SFWM can be written as

$$|\text{II}\rangle \sim |\omega_{-1}\rangle_s |\omega_1\rangle_i + |\omega_{-2}\rangle_s |\omega_2\rangle_i + \dots, \quad (24)$$

where $|\omega\rangle_q \equiv |1\rangle_{q,\omega}$ and the multimode spectral emission is made explicit. Equation (24) is an entangled state in frequency due to the energy conservation ($2\omega_p = \omega_{-1} + \omega_1 = \omega_{-2} + \omega_2 = \dots$). If the herald photon, e.g., the signal, is measured without resolving its frequency, the heralded photon will result in the mixed state $\{|\omega_1\rangle_i; |\omega_2\rangle_i; \dots\}$. On the contrary, if the two-photon state is band-pass filtered to detect only one frequency mode, e.g., the factorizable state $|\text{II}\rangle \sim |\omega_{-2}\rangle_s |\omega_2\rangle_i$, the heralding detection will project the

TABLE I. Relevant metrics to characterize heralded single photon sources. For the meaning of the different symbols, we refer to the main text following the quoted equations.

Parameter	Expression	Equations
Purity (P)	$P = g^{(2)}(0) - 1 = R_{ss}(0)R_p/R_sR_s(\infty)$	Eq. (38)
Join spectral amplitude (JSA)	$F(\omega_s, \omega_i) = \int_{-\infty}^{+\infty} d\omega \alpha(\omega) \alpha(\omega_s + \omega_i - \omega) \phi(\omega_s, \omega_i, \omega)$	Eq. (40)
Antibunching dip	$g_h^{(2)}(0) = R_{ssi}(0)R_i/R_{si}^{(1)}(0)R_{si}^{(2)}(0)$	Eq. (51)
Heralding efficiency	$\eta_H^{i(s)} = R_{si}/R_{s(i)}\eta_d^{i(s)}$	Eq. (54)
Intrinsic heralding efficiency	$\eta_I^{i(s)} = \eta_H^{i(s)}/T_{i(s)}$	Eq. (55)
Klyshko efficiency	$\eta_K^{i(s)} = R_{si}/R_{s(i)}$	Eq. (53)
Filtered heralded efficiency	$\eta_F = S_{SI}^F/S_{SI}$	Eq. (56)
Coincidence to accidental ratio	$CAR = (R_{si} - R_{acc})/R_{acc}$	Eq. (60)
Brightness	$B = [R_{SP}^{meas}/(\eta_K R_h) - (2 - \eta_K)g_h^{(2)}(0)/2]\eta_{coupl}R_h/R_p$	Eq. (59)
Indistinguishability	$I = N^2 \int d\omega_s d\omega_i F_2^*(\omega_s, \omega_i) F_1(\omega_s, \omega_i) $	Eq. (66)
Heralded HOM visibility	$V_{HH} = (R_{4s}^{max} - R_{4s}^{min})/(R_{4s}^{max} + R_{4s}^{min})$	Eq. (68)

heralded idler photon into the pure state $|\omega_2\rangle_i$. This example shows that only by breaking the frequency correlations, pure heralded single photons can be produced. This does not mean that filtering is the only possibility, but it means that single mode emission is a fundamental requirement for unit purity.

Following the squeezer model of Eq. (21), the multimode character of the state is easily described through the Schmidt decomposition.⁶⁹ This decomposition gives a useful picture to quantify the degree of correlation between signal and idler photons. Any state $|\Psi\rangle$ describing a composite system (e.g., two-photon states) can be decomposed as

$$|\Psi\rangle = \sum_i \sqrt{\lambda_i} |u_i\rangle \otimes |v_i\rangle, \quad (25)$$

where $\{|u_i\rangle\}$ and $\{|v_i\rangle\}$ are orthonormal basis states called Schmidt modes. Each subsystem of $|\Psi\rangle$ is related to one and only one Schmidt mode. The Schmidt coefficients λ_i weight each pair of modes, and the decomposition is such that $\sum_i \lambda_i = 1$. The number of Schmidt modes required to fully describe the state measures its degree of factorizability, i.e., its multimode character. This is quantified through the Schmidt number K as

$$K = \frac{1}{\sum_i \lambda_i^2}. \quad (26)$$

The individual purity of the photons in the pair is then

$$P = \frac{1}{K}. \quad (27)$$

$K = 1$ means a perfectly factorizable two-photons state, which in turn indicates unit purity heralded single photons. If residual correlations are present, $K > 1$ is expected. In the limit of low pumping power, i.e., low squeezing, K can be directly evaluated by measuring $g^{(2)}(0)$ as⁷⁰

$$K = \frac{1}{P} = \frac{1}{g^{(2)}(0) - 1}. \quad (28)$$

The experimental measurement of $g^{(2)}(0)$ is thus a powerful tool to characterize the heralded state purity ($P = g^{(2)}(0) - 1$) or to assess

the number of contributing modes to the state. What does a $g^{(2)}$ measurement of the individual signal or idler beam provide? Considering Eq. (21), the two-photon state is given as a collection of two-mode squeezers. The statistics of the pair state, measured by $g^{(2)}$, depends on the number of squeezers involved, i.e., the number of modes excited. If the state is pure, a single squeezer is involved, thus the emission of the individual twin beams is a single mode. A state described by a single two-mode squeezer, as shown in Eq. (22), is a coherent superposition of photon number correlated Fock states and is characterized by thermal photon statistics in each mode.⁷¹ If the two-photon state, on the contrary, is characterized by an infinite number of squeezers, i.e., the state is maximally entangled, the statistics of the signal or idler beam is Poissonian. This is a consequence of the fact that the convolution of an infinite number of thermal distributions converges to a Poissonian distribution.⁷² Therefore, we expect for a pure heralded single photon ($P = 1$)

$$g^{(2)}(0) = 2, \quad (29)$$

and for a maximally entangled photon pair ($P = 0$)

$$g^{(2)}(0) = 1. \quad (30)$$

Finally, depending on the degree of correlation between the signal and idler photons, the result of a measurement of the degree of second order coherence is expected to be $1 \leq g^{(2)} \leq 2$.

Second order coincidence measurements have been widely used in optics; as an example they have been used to characterize single photon detectors.^{73,74} Here, we deal with the specific case of the correlation within a single beam. In order to measure the $g^{(2)}$, the HBT interferometer is used (see Fig. 2). Note that to inspect the single mode character of the idler/signal photon, only one beam is used without any heralding from the other, as shown in Fig. 4(a). In this case, the $g^{(2)}$ is said to be unheralded. Using the self-correlation to measure the purity has the already mentioned advantage of being insensitive to losses. However, this measure is affected by the background noise. This noise is not relevant for other methods to measure the purity, e.g., in the measure of the joint spectral intensity (JSI) (see Sec. III B).⁷⁵ Moreover, the $g^{(2)}$ method to measure the purity applies for an

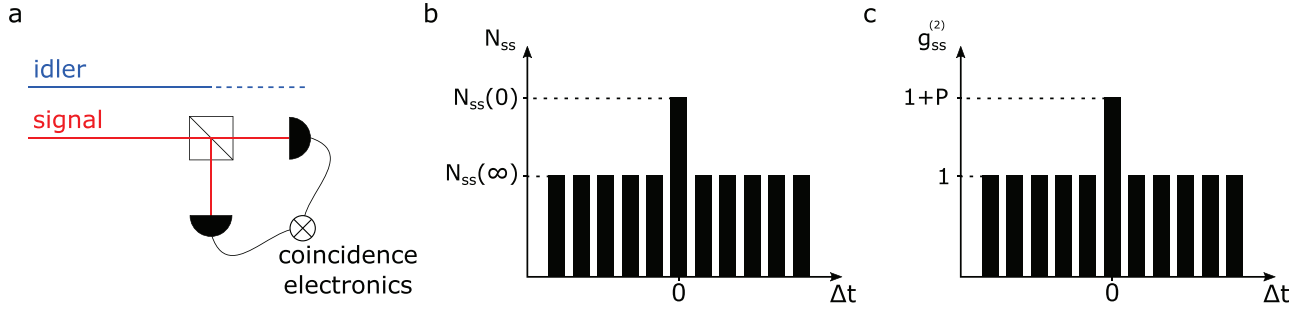


FIG. 4. Purity measurement of the heralded signal photon via unheralded $g^{(2)}(\Delta t)$. (a) HBT setup for the $g^{(2)}(\Delta t)$ measurement of the signal beam. The idler photons are simply discarded. (b) Coincidence counts N_{ss} as a function of the delay Δt between the detections on the two detectors. (c) $g^{(2)}(\Delta t)$ measurement, showing the relationship between purity P and $g^{(2)}(0)$.

integration time window of the detector that is much longer than the pulse duration. The integration time is limited by the internal jitter and the finite gating time of the detector. If the temporal resolution and the pulse duration are comparable in time, the $g^{(2)}$ has to be deconvoluted by the detector time window.⁷⁰ Being the typical measured pulses of the order of pico- or femtoseconds and the detection time resolutions of the order of hundreds of picoseconds, the deconvolution of the $g^{(2)}$ is usually not performed. For a formal treatment of the $g^{(2)}$, considering also the detector integration time, we refer the reader to Ref. 70.

From Eq. (2) and assuming to measure the signal beam (the same analysis can be carried out for the idler), the $g^{(2)}$ can be rewritten in terms of probabilities as

$$g^{(2)}(\Delta t) = \frac{P_{ss}(\Delta t)}{P_s P_s}, \quad (31)$$

where $P_{ss}(\Delta t)$ is the probability of a coincidence detection at a fixed delay Δt and P_s is the probability of a signal detection on one of the two detectors. Notice that the time delay is not explicit in P_s , since $P_s(t) = P_s(\Delta t)$ is a constant probability. Equation (31) derives from the definition of the correlation function, which measures the ensemble average of a quantity that is correlated. In this specific case, with the creation $\hat{a}^\dagger(t)$ and destruction $\hat{a}(t)$ operators at time t , the numerator in Eq. (2) cross correlates the detection events in the two detectors of the HBT. In this way, it measures the number of total coincidences, per each Δt , averaged over all the possible coincidence events, i.e., the total number of pump pulses (for the pulsed case). Therefore, we can write

$$\langle \hat{a}^\dagger(t) \hat{a}^\dagger(\Delta t) \hat{a}(\Delta t) \hat{a}(t) \rangle = \frac{N_{ss}(\Delta t)}{N_p}, \quad (32)$$

with N_{ss} being the number of signal coincidences measured at a single value of Δt and N_p being the total number of pump pulses. We are considering the pulsed case, since it is more intuitive. The extension to the CW case is straightforward. The number of pump pulses is the number of possible coincidence events, because the parametric process has a finite probability to generate the photon pair per each pulse, such that $P_{ss} = N_{ss}/N_p$ and $P_s = N_s/N_p$, where N_s is the number of signal counts on one detector. By introducing the rates R , which are more accessible experimental quantities, we have

$$N_{ss} = R_{ss} T_{meas}, \quad (33a)$$

$$N_s = R_s T_{meas}, \quad (33b)$$

$$N_p = R_p T_{meas}, \quad (33c)$$

where R_{ss} is the rate (in counts/s) of coincidence counts, R_s is the rate of single counts, R_p is the pump rate, and T_{meas} is the integration time of the measurement. Thus, by using Eq. (32), Eq. (31) is rewritten for the pulsed case as

$$g^{(2)}(\Delta t) = \frac{R_{ss}(\Delta t)}{R_s R_s} R_p, \quad (34)$$

while for the CW case, Eq. (34) is⁷⁶

$$g^{(2)}(\Delta t) = \frac{R_{ss}(\Delta t)}{R_s R_s} \tau_b^{-1}, \quad (35)$$

where τ_b is the time window where coincidences are counted. In principle, the coherence time of the process τ_c should be used instead of τ_b in Eq. (35); however, since usually $\tau_c \ll \tau_b$, the actual time interval in which a coincidence may occur is τ_b . From an experimental point of view, $g^{(2)}(0)$ can be simply evaluated by measuring the single and the coincidence rates at $\Delta t = 0$. An equivalent approach is based on the measurement of the coincidence counts $N_{ss}(\Delta t)$ only. Considering the pulsed case, the coincidence counts at zero and the nonzero delay can be expressed as

$$N_{ss}(0) = P_{ss}(0) N_p, \quad (36a)$$

$$N_{ss}(\Delta t > 0) = P_s P_s N_p, \quad (36b)$$

where $P_{ss}(\Delta t > 0) = P_s P_s$ because, when the time delay between coincidence counts is longer than the pulse duration, the clicks from the two detectors are independent events. Now the following relationship holds:

$$\frac{N_{ss}(0)}{N_{ss}(\Delta t > 0)} = \frac{P_{ss}(0) N_p}{P_s P_s N_p} = \frac{P_{ss}(0)}{P_s P_s} = g^{(2)}(0). \quad (37)$$

Notice that both Eqs. (37) and (11) give an expression for $g^{(2)}(0)$: while Eq. (37) uses coincidence and single detection probabilities, Eq. (11) uses photon number probabilities. Therefore, the two expressions are not directly comparable. The reasoning used to obtain Eq. (37) applies to the CW case too. Therefore, $g^{(2)}(0)$, and thus the purity, can

be easily measured as the ratio between the nondelayed and the delayed coincidence counts. In other terms, the $g^{(2)}(\Delta t)$ is obtained by normalizing $N_{ss}(\Delta t)$ with $N_{ss}(\Delta t > 0)$ in the pulsed case or $N_{ss}(\infty)$ in the CW case [Fig. 4(b)]. From the measured $N_{ss}(\Delta t)$, $g^{(2)}(\Delta t)$ can be derived [Fig. 4(c)], and

$$P = g^{(2)}(0) - 1 = \frac{N_{ss}(0)}{N_{ss}(\infty)} - 1. \quad (38)$$

B. Joint spectral intensity

The two-photon state for SPDC or SFWM processes can be written as⁷⁷

$$|II\rangle \sim \iint d\omega_s d\omega_i F(\omega_s, \omega_i) \hat{a}_s^\dagger(\omega_s) \hat{a}_i^\dagger(\omega_i) |0\rangle_s |0\rangle_i, \quad (39)$$

where $|\omega\rangle_q \equiv |1\rangle_{q,\omega} = \hat{a}_q^\dagger(\omega)|0\rangle$ represents one photon in mode q with frequency ω . $F(\omega_s, \omega_i)$ is the joint spectral amplitude (JSA), also named as the normalized biphoton wavefunction. Note that the JSA only represents the biphoton wavefunction in the low squeezing limit. In this case, the JSA embeds the information on the bandwidth of the pump, signal, and idler photons, and it is given by

$$F(\omega_s, \omega_i) = \int_{-\infty}^{+\infty} d\omega \alpha(\omega) \alpha(\omega_s + \omega_i - \omega) \phi(\omega_s, \omega_i, \omega), \quad (40)$$

where $\alpha(\omega)$ is the complex amplitude of the pump spectrum centered at ω_p and

$$\phi(\omega_s, \omega_i, \omega) = \exp\left(\frac{i\Delta\beta L}{2}\right) \text{sinc}\left(\frac{\Delta\beta L}{2}\right), \quad (41)$$

is the phase matching (PM) function, with L being the waveguide length and $\Delta\beta$ being the phase mismatch parameter [Eq. (16)].⁷⁸ The JSA quantifies the probability density for the generation of the first photon of the pair in the ω_s mode and the second photon in the ω_i mode. From now on, we will consider only the case of SFWM, without loss of generality. The modulus square of the JSA is called the joint spectral intensity (JSI). In the JSA, the complex phase is explicit, while in the JSI, it is not. The JSI is the commonly measured quantity, since it depends on the field intensity. The JSI is used to characterize the degree of correlation between the signal and idler photons.⁷⁹ In fact, since the JSI takes into account all the frequencies and bandwidths of the involved waves (pump, signal, and idler), it directly reflects the spectral distribution of the signal (idler) once the frequency of the idler (signal) is given. The JSI quantifies the degree of correlation, according to the Schmidt decomposition. In fact, the JSA can be decomposed according to Eq. (25)⁸⁰

$$F(\omega_s, \omega_i) = \sum_n \lambda_n u_n(\omega_s) v_n(\omega_i), \quad (42)$$

with u_n and v_n being two complete sets of orthonormal functions. If the state is pure, then the JSA should be factorizable. Among the many separable complex-valued functions of two variables, a two-dimensional Gaussian can be simply used to represent JSA. The factorizable condition corresponds to a single mode emission, i.e., $F(\omega_s, \omega_i) \sim u_n(\omega_s) v_n(\omega_i)$ in the Schmidt decomposition picture. If correlations are present, the JSA is not factorizable.

The decomposition in Eq. (42) is not possible with the JSI. From the measurement of the JSI, it is not possible to derive directly the JSA, since one needs to measure the phase of the optical beam.^{80–82} Methods to retrieve the joint spectral phase have been proposed, such as the stimulated emission tomography.⁸³ Recently another method has been proposed, which allows us to directly retrieve the joint spectral phase to reconstruct the complex JSA in a PIC.⁸² These methods are more demanding than the direct measure of the JSI. Therefore, one usually considers a singular value decomposition of the square root of the JSI. This provides a lower bound for the Schmidt number, since it does not account for the phase correlations.⁷⁵ From the Schmidt number, it is possible to calculate the purity of the heralded single photon as $P = 1/K$.⁸⁴ As a consequence, the JSI provides only an upper bound to the purity.⁷⁵ Because of this, the JSI measurement is typically reported together with the $g^{(2)}$ measurement of the purity: the first quantity provides a first estimation of the purity and the joint spectral information of the source and the second measures the purity. When measuring the JSI, the shape is expected to range from a thin line (perfect correlation) to a circular shape (perfect decorrelation),⁸⁵ as shown in Fig. 5.

Considering the case in which no phase correlations are present, the circular shape case directly resembles the case in which the two-photon system is in a separable state. This means maximum purity, $P = 1$. When the JSI is circular, the JSA can be factorized, such that $F(\omega_s, \omega_i) = F_s(\omega_s) F_i(\omega_i)$.²¹ The case of a separable bi-photon wavefunction is exactly the pursued condition for the heralded single photon operation, where the maximum purity for the single photon state can be attained only by eliminating the correlation between the twin photons.²¹

The simplest technique to get uncorrelated pairs in a SFWM experiment is the spectral filtering. Narrow bandpass filters are used on both the herald and heralded beams, thus shaping the JSA by removing the unwanted frequency modes.⁸⁶ From the Schmidt decomposition point of view, filtering means selecting only certain Schmidt modes, approaching single mode emission and, thus, the factorizability of the state. The filtering process is sketched in Fig. 6(a), where δ_s and δ_i are the filter bandwidths for the signal and idler beams, respectively. The effect of the filters can be included in the description of the bi-photon wavefunction by multiplying $F(\omega_s, \omega_i)$ in Eq. (40) by two Gaussian functions of the form

$$f_j(\omega_j) = \frac{1}{\Gamma} \exp\left[-\frac{(\omega_{j0} - \omega_j)^2}{\delta_j^2} 4 \ln(2)\right], \quad (43)$$

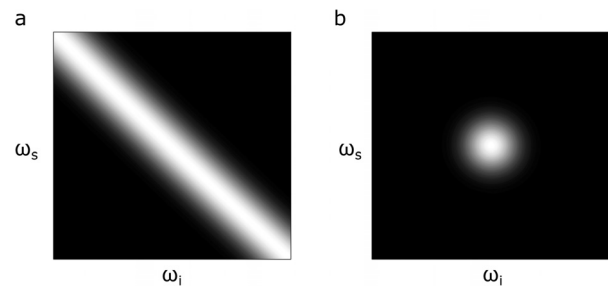


Fig. 5. Joint Spectral Intensity (JSI) for (a) correlated and (b) decorrelated photon pairs. ω_s and ω_i are the signal and idler frequencies, respectively.

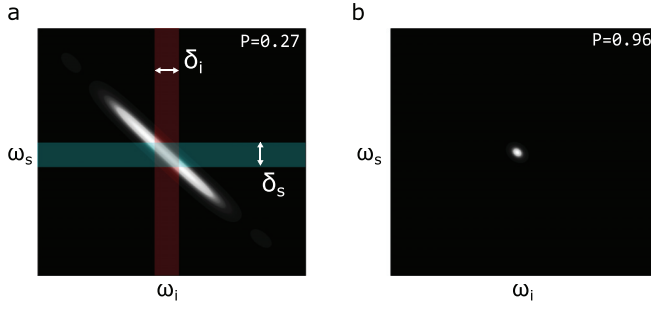


FIG. 6. (a) Unfiltered JSI, which yields a $P = 0.27$. The colored stripes represent the action of filters on the JSI. δ_i and δ_s are the bandwidths of the idler and signal filters, respectively. (b) Filtered JSI, with a purity increased to 0.96.

with Γ being a normalization factor, ω_{j0} being the central filter frequency, δ_j being the filter bandwidth, and $j = i, s$ labeling the idler and signal, respectively. Filtering manipulates the JSA, achieving high purity single photon states, as shown in Fig. 6(b). However, filters lower the brightness and the heralding efficiency of the source, since photons occupying the unwanted modes are irredeemably lost.

A more efficient solution resides on the engineering of the JSA. Equation (40) shows that JSA depends on the phase matching and on the pump profiles. By properly engineering such functions, the JSA can be made intrinsically uncorrelated, such that only one mode of the Schmidt decomposition can be populated. This removes the needs of filters, which improves the source brightness.⁸⁷ Other schemes rely on engineering the pump pulse.^{88,89}

The purity upper bound of a heralded single photon source can be calculated by means of the JSI as⁹⁰

$$P = \int \int d\omega d\omega' |q_s(\omega, \omega')|^2, \quad (44)$$

where $q_s(\omega, \omega') = \int d\omega'' F(\omega'', \omega) F^*(\omega'', \omega')$, with q_s related to the density matrix of the heralded signal photon as

$$\rho_s = \text{Tr}_i\{|II\rangle\langle II|\} = \int \int d\omega d\omega' q_s(\omega, \omega') \hat{a}_s^\dagger(\omega) |0\rangle\langle 0| \hat{a}_s(\omega'). \quad (45)$$

Experimentally, the JSI can be measured directly by sampling with narrow-band filters the signal–idler coincidences over the whole

generation bandwidth⁷⁹ or by using a dispersion fiber-based spectrometer.⁹¹ To speed up the measurements, it is possible to infer the JSI for the spontaneous process by a stimulated process.⁸³ The principle here is to stimulate the process by adding a seed wave which spans the entire spectral bandwidth of the signal or idler waves. The generated stimulated photon pairs show the same correlations as the spontaneous ones. This means that the reconstructed stimulated JSI resembles the spontaneous one. The spectral correlations in the stimulated experiments can be resolved with a better signal to noise ratio and resolution than the ones in the spontaneous experiments. A further advantage of the stimulated tomography is that it can also resolve the joint spectral phase.⁸¹ Indeed, the idler and signal can be correlated through a phase relationship which detrimentally affects the state purity. Through the spontaneous measurement of the JSI, such a phase relationship cannot be resolved, and the measured purity sets just an upper bound to the actual purity. On the contrary, the $g^{(2)}$ measurement of the purity takes into account the phase information.

C. Antibunching dip

The definition of purity for heralded photons has a different meaning with respect to atom-like emitters. With atom like emitters, in fact, the $g^{(2)}$ exhibits a clear antibunching dip at zero delay, caused by the intrinsic single photon emission character of the source. This is not the case for squeezed states generated through parametric processes. In fact, as shown in Sec. III A, each isolated twin-beam shows a peak at $g^{(2)}(0)$, being super-Poissonian. The sub-Poissonian behavior is recovered only when heralding is applied. Let us assume the idler photon as the herald photon and the signal photon as the heralded photon. When heralding is applied, the second order correlation of the signal is conditioned to the detection of the herald, thus changing the expression of the $g^{(2)}$. The conditioned second order correlation function $g_h^{(2)}$ is defined as the self-correlation of the signal beam triggered by the detection of its twin idler.

The experimental setup to measure the conditioned $g_h^{(2)}$ is shown in Fig. 7(a). It is based on a HBT interferometer with the idler (herald) arm added in the timing electronics (correlator, start/stop electronics, etc.). The detection of the herald triggers the acquisition of the signal–signal coincidences. In this sense, the resulting $g_h^{(2)}$ is conditioned. Starting from the second order coherence formalism of Eq. (2), we derive $g_h^{(2)}$

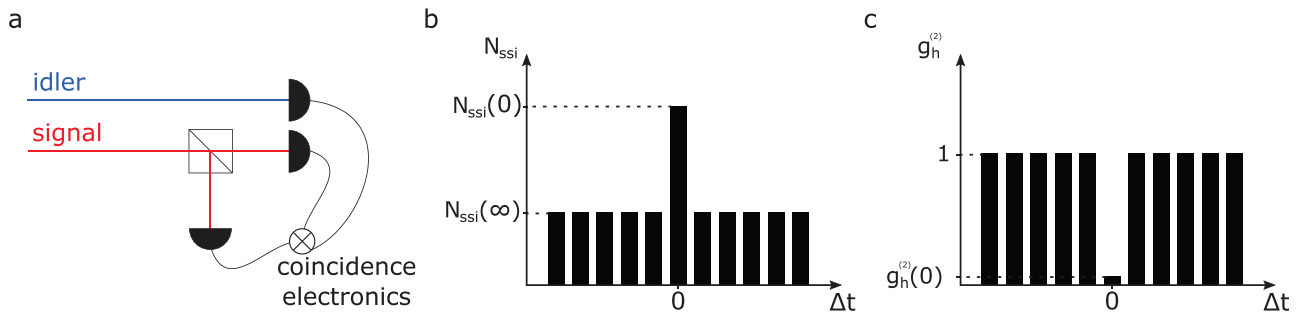


FIG. 7. Measurement of the heralded $g_h^{(2)}(\Delta t)$. (a) HBT setup for the $g_h^{(2)}(\Delta t)$ measurement when the herald photon is the idler photon and the heralded one is the signal photon. The idler photons trigger the coincidence electronics. (b) Threefold coincidence counts $N_{ssi}(\Delta t)$. Only one of the two detectors of the signal photons has a delay Δt with respect to the detector for the idler photon. $\Delta t = t_2 - t_1$, where $t = t_1$ is the time at which the idler detector and one of the two signal detectors click, while t_2 is the time at which the second detector on the signal clicks. (c) $g_h^{(2)}(\Delta t)$ computed according to Eq. (50).

$$g_h^{(2)}(t_1, t_2 | t_i) = \frac{\langle \hat{a}_1^\dagger(t_1) \hat{a}_2^\dagger(t_2) \hat{a}_2(t_2) \hat{a}_1(t_1) \rangle_{pm}}{\langle \hat{a}_1^\dagger(t_1) \hat{a}_1(t_1) \rangle_{pm} \langle \hat{a}_2^\dagger(t_2) \hat{a}_2(t_2) \rangle_{pm}}, \quad (46)$$

$\hat{a}_j^\dagger(t), \hat{a}_j(t) (j = 1, 2)$ are, respectively, the creation and destruction operators for a photon in the signal mode at time t , 1 and 2 refer to the two detectors, and $\langle \cdot \rangle_{pm}$ is the average over the post-measurement state upon the detection of the herald idler photon at time t_i .⁹² Averaging over the post-measurement state means that only events with a detected idler photon have to be considered, namely,

$$\langle \hat{X} \rangle_{pm} = \frac{\langle \hat{a}_i^\dagger(t_i) \hat{X} \hat{a}_i(t_i) \rangle}{\langle \hat{a}_i^\dagger(t_i) \hat{a}_i(t_i) \rangle}, \quad (47)$$

with $\hat{X}$ being a general operator. According to Eq. (47), Eq. (46) can be rewritten, making explicit the creation/annihilation operators of the idler ($\hat{a}_i^\dagger \hat{a}_i$), such that

$$g_h^{(2)}(t_1, t_2 | t_i) = \frac{\langle \hat{a}_i^\dagger \hat{a}_1^\dagger \hat{a}_2^\dagger \hat{a}_1 \hat{a}_2 \hat{a}_i \rangle}{\langle \hat{a}_i^\dagger \hat{a}_1^\dagger \hat{a}_1 \hat{a}_i \rangle \langle \hat{a}_i^\dagger \hat{a}_2^\dagger \hat{a}_2 \hat{a}_i \rangle} \langle \hat{a}_i^\dagger \hat{a}_i \rangle, \quad (48)$$

where the temporal dependence has been omitted for simplicity. Typically, the measurement is prepared with the idler arriving on its detector with zero delay with respect to the signal on (e.g.,) detector 1. This means that $t_i = t_1 = t$. Therefore, the clicks by the second signal detector are expressed as delayed by $\Delta t = t_2 - t$ with respect to the other two detections. $g_h^{(2)}$ becomes a function of Δt only, and, according to Eq. (32), it can be written as

$$g_h^{(2)}(t, t + \Delta t | t) \equiv g_h^{(2)}(\Delta t) = \frac{P_{ssi}(\Delta t)}{P_{si}^{(1)}(t) P_{si}^{(2)}(\Delta t)} P_i(t), \quad (49)$$

with $t > 0$, P_{ssi} being the probability of threefold coincidences, and $P_{si}^{(1)}, P_{si}^{(2)}$ being the signal-idler coincidences with respect to the first or the second signal detector. Starting again from Eq. (32), it is possible to express $g_h^{(2)}$ as a function of the number of coincidences

$$g_h^{(2)}(\Delta t) = \frac{N_{ssi}(\Delta t)}{N_{si}^{(1)}(t) N_{si}^{(2)}(\Delta t)} N_i(t), \quad (50)$$

with N_{ssi} being the threefold coincidences between the three detectors, $N_{si}^{(1)}$ and $N_{si}^{(2)}$ being the twofold coincidences between the signal and idler, and N_i being the number of idler detections. A similar expression can be written with the coincidences rates

$$g_h^{(2)}(\Delta t) = \frac{R_{ssi}(\Delta t)}{R_{si}^{(1)}(t) R_{si}^{(2)}(\Delta t)} R_i(t), \quad (51)$$

where R_{ssi} is the three fold coincidence rate and R_{si} is the rate of coincidences between the idler and the signal.

Equations (50) and (51) hold for both CW and pulsed operation. The quantity of interest is the zero-delay $g_h^{(2)}(0)$. Therefore, it is not sufficient to monitor the threefold coincidences only, which do not exhibit any dip at zero delay, see Fig. 7(b). Threefold, twofold, and herald singles have to be measured simultaneously and, by Eq. (50), the value of $g_h^{(2)}(0)$ can be derived, as shown in Fig. 7(c).

The act of heralding can be seen as a procedure to nullify the probability of detecting zero signal photons, i.e., to force $P(0) = 0$. The signal beam, in fact, exhibits a super-Poissonian statistics with a very low number of pairs per pulse. This is the typical condition for

the single photon operation.⁴⁸ Hence, the signal beam statistics satisfies $P(0) \gg P(1) \gg P(2) \dots$, as shown in Fig. 1(c). As soon as the herald trigger is applied, the signal detectors look at the presence of photons only when the time correlated photons have been generated, thus minimizing $P(0)$, as shown in Fig. 1(d). This in turn increases the average photon number per pulse $\mu \simeq P(1) \gg P(2)$.²¹ Therefore, according to Eq. (11),

$$g_h^{(2)}(0) \simeq \frac{2P(2)}{\mu^2} \ll 1. \quad (52)$$

Here, the single photon sub-Poissonian statistics emerges and $g_h^{(2)}(0)$ is closer and closer to zero as the pump power is reduced.

D. Heralding efficiency

In order to satisfy the high purity requirement, heralded sources are typically heavily filtered. This affects the brightness of the heralded single photon source, because the filtered photons are irredeemably lost. Also, the losses on the heralding channel are detrimental, because the rate of usable single photons depends on the rate of successful heralding events. To have a successful heralding event, both the herald and heralded photons have to incur minimum losses along the measurement channels. Therefore, one important parameter that characterizes the practical feasibility of a heralded single photon source is the heralding efficiency η_H . η_H quantifies the probability that, given the detection of the herald photon, the heralded photon will also be detected.⁹³ In this simple definition, η_H is known as the Klyshko efficiency, η_K ,^{74,94} and can be measured by

$$\eta_K^{s(i)} = \frac{R_{si}}{R_{i(s)}}, \quad (53)$$

where R_{si} is the rate of coincidences between the signal and idler photons and $R_{i(s)}$ is the rate of the detected idler (signal) photons.

A preferred definition of the heralding efficiency considers the detection efficiency of the detector η_d . In this case, the heralding efficiency with the idler (signal) as the herald is measured as

$$\eta_H^{s(i)} = \frac{R_{si}}{R_{i(s)} \eta_d^{s(i)}}, \quad (54)$$

where $\eta_d^{s(i)}$ is the detection efficiency of the signal (idler) detector. The presence of η_d allows us to decouple the quality of the overall source from the available detectors.

Additionally, the intrinsic heralding efficiency η_I is used.⁹⁵ η_I considers the efficiency in the heralding process removing those losses which only depend on the quality of the instrumentation involved, such as filter losses, detector efficiencies, coupling losses and, in general, all the losses from the source to the detectors. In this way, only the losses intrinsic to the photon source are accounted for. The intrinsic heralding efficiency is evaluated as

$$\eta_I^{s(i)} = \frac{\eta_H^{s(i)}}{T_{s(i)}}, \quad (55)$$

where $T_{s(i)}$ is the transmission of the signal (idler) by the photon source. Notice that if filters are used to increase the purity of the source, the losses induced by those filters have to be considered, being intrinsically a component of the whole source. In this sense, T has to

account only for the insertion losses of the filters, so that they can be considered with unit transmittivity at their peak wavelength.

Since filtering is used to purify the single photon state, a related efficiency is introduced to compare different filtered sources. This is named as the filtered heralded efficiency η_F . η_F quantifies the probability that, given that the herald has passed its filter, the heralded photon will pass the corresponding filter.⁹⁶ It intuitively quantifies the effect of filters on the heralding, while neglecting the role of the other parameters. Starting from the JSI,

$$\eta_F = \frac{S_{JSI}^F}{S_{JSI}}, \quad (56)$$

where S_{JSI} and S_{JSI}^F are, respectively, the areas of the unfiltered JSI and of the filtered one. In other terms, η_F quantifies the amount of JSI, i.e., of the spectral emission, that passes the filters without being altered. The most promising systems to guarantee both high heralding efficiency and purity in a PIC are based on microring resonators and intermodal phase matching, which can leverage their discrete band emission to shape the JSI without external filtering. We will discuss these in Sec. IV.

The different heralding efficiencies discussed in this section scale as

$$\eta_K < \eta_H < \eta_I < \eta_F. \quad (57)$$

E. Brightness

Fundamental for the feasibility of the source is the source brightness B , which is the emission rate of single photons. This parameter is equivalently used for deterministic atomic emitters and heralded sources, but the latter are also characterized by the heralding efficiency. Therefore, for a heralded single photon source, the brightness is directly linked to the pair emission probability. Since the probability of emission scales with the pump power P_p (for a SPDC source, we have $P(1) \propto P_p$, $P(2) \propto P_p^2$ while for SFWM $P(1) \propto P_p^2$, $P(2) \propto P_p^4$), the emission of multiphoton states increases with the pump power. Thus, there is a compromise to be made between high brightness and high single photon fidelity, i.e., $g_H^{(2)} \simeq 0$, for a heralded single photon source. This is not the case for a deterministic source. In this scenario, heralded sources could benefit from PNR detectors, which can select the single pair states among multipair ones. This allows decoupling the brightness from the single photon fidelity, achieving a maximum single photon heralding probability of 25%.⁹⁷ However, PNR detectors are still far from being the standard, therefore the state-of-the-art in heralded single photon sources is still dominated by the trade-off between brightness and single photon fidelity.

If we consider Eq. (22) and limit to the first terms,

$$|II\rangle = \sqrt{P(0)}|0\rangle_s|0\rangle_i + \sqrt{P(1)}|1\rangle_s|1\rangle_i + \sqrt{P(2)}|2\rangle_s|2\rangle_i + \dots, \quad (58)$$

we see that with a nonresolving photon number detector, both single photon and multiphoton terms contribute to the rate of the heralded single photon detections R_{SP}^{meas} . R_{SP}^{meas} is defined as the rate of heralded photons measured on the detector. Following Ref. 97, we assume that only $P(1)$ and $P(2)$ are non-negligible (i.e., $P(2) \gg P(n > 2)$, which is usually the case). Then, we compute the brightness B by subtracting

the contribution of multipair emission to R_{SP}^{meas} . Multipair emission is estimated from the measured $g_H^{(2)}(0)$. Then, if we define the brightness as the probability of generating a heralded single photon and of collecting it in a single mode fiber, we have⁹⁷

$$B = \left[\frac{1}{\eta_K} \frac{R_{SP}^{meas}}{R_h} - (2 - \eta_K) \frac{g_H^{(2)}(0)}{2} \right] \frac{R_h}{R_p} \eta_{coupl}, \quad (59)$$

with η_{coupl} being the coupling efficiency to the single mode fiber, R_h being the heralding rate, and R_p being the pump rate. Equation (59) considers only measurable quantities, providing an easy calculation of the brightness.

F. Coincidence to accidental ratio

To estimate the noise in the heralded single photon source, the coincidence to accidental ratio (CAR) parameter is used. This describes the overall coincidence detection efficiency with respect to spurious coincidence counts. In fact, losses on the idler or signal detection channels cause the occurrence of coincidences between the detection of signal/idler photons and noisy clicks or detection of photons which are not part of a pair. Only the coincidence counts between the signal and idler photons of the same pair are coincidences which characterize the temporal correlation of the emitted pairs. Other apparent coincidence detections of photons of distinct pairs are spurious coincidences. These, together with those coming from noise, are named accidental coincidences. An efficient source of photon pairs is characterized by a rate of good coincidences much greater than the accidental ones. Therefore, the CAR parameter can be defined as^{48,98}

$$CAR = \frac{R_{si} - R_{acc}}{R_{acc}}, \quad (60)$$

with R_{acc} being the accidental coincidence rate. This parameter is closely related to the signal-idler cross-correlation $g_{si}^{(2)} \cdot g_{si}^{(2)}$ can be computed as

$$g_{si}^{(2)}(\Delta t) = \frac{\langle \hat{a}_s^\dagger(t + \Delta t) \hat{a}_i^\dagger(t) \hat{a}_i(t) \hat{a}_s(t + \Delta t) \rangle}{\langle \hat{a}_s^\dagger(t + \Delta t) \hat{a}_s(t + \Delta t) \rangle \langle \hat{a}_i^\dagger(t) \hat{a}_i(t) \rangle} = \frac{R_{si}(\Delta t)}{R_s R_i} R_p, \quad (61)$$

with R_s being the signal photon detection rate, R_i being the idler photon detection rate, and R_p being the pump repetition rate. For a CW pump, Eq. (61) changes accounting for the coincidence time window τ_b as

$$g_{si}^{(2)}(\Delta t) = \frac{R_{si}(\Delta t)}{R_s R_i \tau_b}. \quad (62)$$

Experimentally, the CAR measurement is carried out by connecting the idler and signal detectors to a correlator or to a start/stop timing electronics, as sketched in Fig. 8(a). Considering a pulsed pump, the number of total coincidences C_{tot} measured in the experiment corresponds to the coincidence bin at $\Delta t = 0$, shown in Fig. 8(b). The coincidence bin should be large enough to account for the jitter of the detectors. For a nonzero Δt , the coincidences are accidental, with a total measured number C_{acc} . The resulting CAR is calculated according to Eq. (60)

$$CAR = \frac{C_{si}(0) - C_{si}(\Delta t > 0)}{C_{si}(\Delta t > 0)} = \frac{C_{tot} - C_{acc}}{C_{acc}}, \quad (63)$$

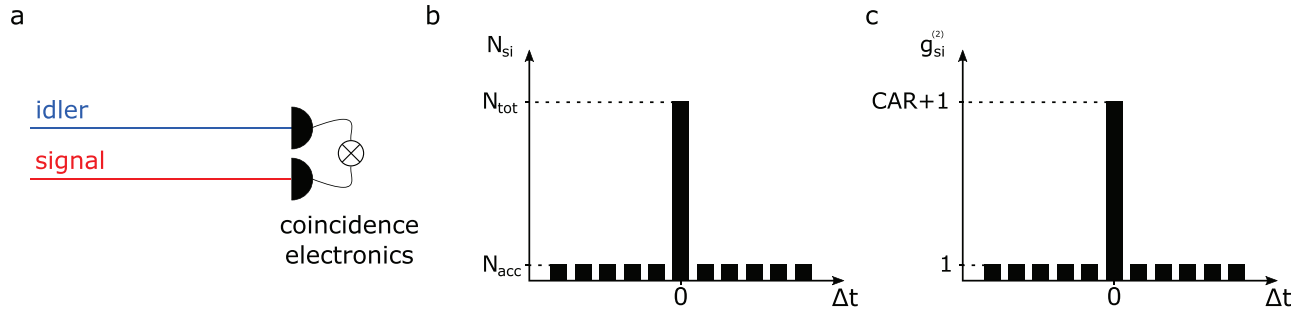


Fig. 8. (a) Setup for the CAR measurement. The detectors for the idler and signal photons are connected to the coincidence electronics. (b) Coincidence counts as a function of the delay Δt between the idler and signal detections. At $\Delta t = 0$, the total simultaneous coincidences N_{tot} are measured, while at nonzero delays, the accidental counts N_{acc} are reported. (c) $g_{\text{si}}^{(2)}(\Delta t)$ between the signal and idler. The experimental relationship of $g_{\text{si}}^{(2)}(\Delta t)$ with the CAR is visible at $\Delta t = 0$.

with $C_{\text{si}} = R_{\text{si}} T_{\text{meas}}$, where T_{meas} is the measurement time. In Eq. (63), Δt should be an integer multiple of the repetition rate in the pulsed case,⁹⁹ while $\Delta t \gg 0$ is used for the CW case.¹⁰⁰ In the pulsed case, when $\Delta t > 0$, the coincidence probability between the signal and idler can be factorized in the product of the individual detections, the events being independent when not belonging to the same pump pulse. As a consequence, the rate of accidentals can be rewritten as

$$R_{\text{acc}} = R_{\text{si}}(\Delta t > 0) = \frac{R_{\text{s}} R_{\text{i}}}{R_{\text{p}}}. \quad (64)$$

Inserting Eqs. (64) and (62) into Eq. (60), we obtain

$$\text{CAR} = g_{\text{si}}^{(2)}(0) - 1, \quad (65)$$

which links the $g_{\text{si}}^{(2)}$ measurement to the CAR parameter, as sketched in Fig. 8(c). The CAR is therefore a fundamental parameter to characterize the quality of a source. The CAR is analogous to the signal to noise ratio in a generic photon source. Since the CAR depends on the detection efficiency, jitter, coincidence window, losses, and noise, it is a parameter that not only characterizes the source but also depends on the experimental setup. This means that the same source measured in different setups could show different CAR values.

G. Indistinguishability

Purity is a property that directly characterizes the individual single photon state. It does not depend on the presence of other photons or on the interaction with other single photons. However, since in quantum information we manipulate photons, we are interested that any heralded single photon is indistinguishable to its contemporaries. This motivates the interest to the purity of single photon states and to their indistinguishability. Indeed, several quantum protocols and applications require a high visibility interference between single photons. Interference needs high purity and high indistinguishability between the single photons. Two photons are indistinguishable when they are identical with respect to all their degrees of freedom, even if they have been produced by two different sources. Polarization and momentum indistinguishability are easily satisfied with heralded single photons, but frequency indistinguishability can be an issue. In fact, the spectral emission properties of a single or of many sources are governed by the phase matching between the source pump photons. Phase matching in a PIC is highly sensitive to fabrication

imperfections, which occur almost randomly. State-of-the-art processing technologies achieve at most a dimension resolution of few nm, while other sources of fluctuations in the waveguide parameters could be composition variations, thickness variations, surface contamination, etc.¹⁷ These affect the local refractive indices which in turn affect the effective mode indices, yielding a disorder induced random shift of the phase matching frequencies. As a result, two nominally identical sources generate distinguishable photons by uncontrolled fabrication variations. Note that here indistinguishability neither takes into account the purity of the single photon state nor heralding is considered. Indistinguishability differs from the visibility of the quantum interference between two photons generated by heralded single photon sources, as we will see in Sec. III H.

From an experimental point of view, indistinguishability can be measured by a Hong–Ou–Mandel (HOM) interferometer or by the setup shown in Fig. 9(a). This is based on two nominally identical single photon sources [Fig. 9(a)] either on the same chip or on different chips. The sources generate two pairs of signal–idler photons. The idler and signal photons are routed in order to input the two idler photons in a Mach–Zehnder Interferometer (MZI) and the two signal photons in another MZI. Only two outputs, one for each MZI, are used to record coincidence counts. Notably, the phase of the photons are controlled to change the interference in the measuring channels. Heaters [orange rectangles shown in Fig. 9(a)] are used to change the phase. To measure the indistinguishability, the setup can be used to implement three different techniques: JSI overlap, reversed Hong–Ou–Mandel (HOM) interference, and quantum state tomography. We consider the case study of Ref. 95, where the three techniques are used to characterize the same kind of integrated sources.

1. Joint spectral intensity overlap

Characterizing the spectral indistinguishability of two sources means measuring and comparing their spectral emissions. The JSI, being a tool to evaluate the spectral correlations within the same pair of photons, represents exactly the spectral properties of the source. If two sources are identical in frequency, they must exhibit the very same JSI.¹⁰¹ Therefore, the overlap integral between the JSIs of different sources ranges from 0 (distinguishable) to 1 (indistinguishable). The experiment is carried out by operating the two sources independently. To reconstruct each JSI, two tunable narrow band filters (one per each channel) are placed before the signal and idler

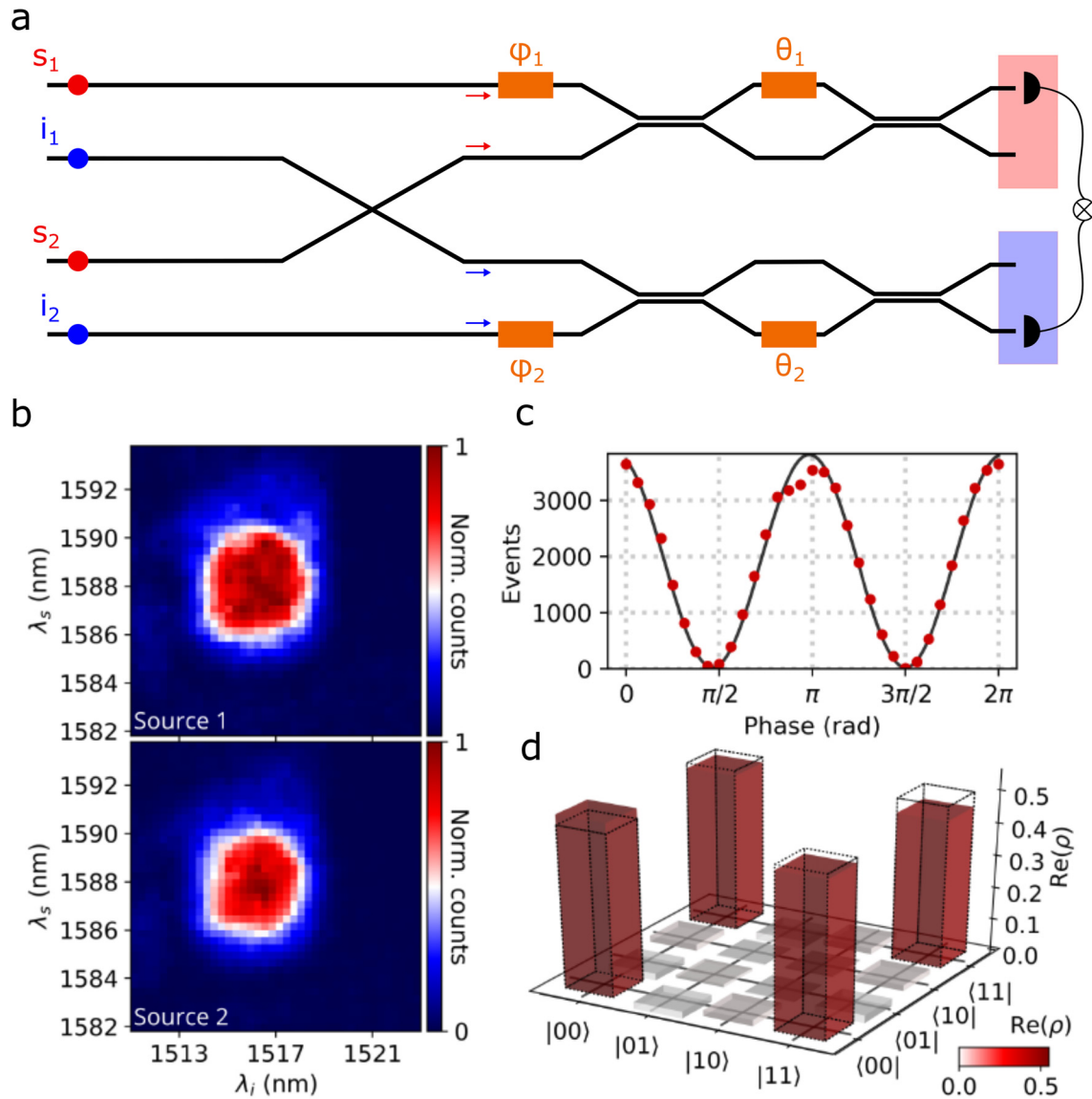


FIG. 9. (a) Experimental setup for indistinguishability measurements. (b) Measured JSI for two nominally identical SFWM sources. (c) Reversed HOM interference. The indistinguishability corresponds to the visibility of the interference fringes that is 0.987(2) here. (d) Quantum state tomography of the two-qubit entangled state. The transparent bars represent the maximally entangled state $|\Phi_+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ used for comparison. All figures reprinted with permission from Paesani *et al.*, Nat. Commun. 11, 1 (2020) (Ref. 95). Copyright 2020 Authors, licensed under a Creative Commons Attribution 4.0 International License.

detectors. As an example, shown in Fig. 9(b), the JSIs of two nominally identical SFWM sources are reported: the normalized overlap integral between these two JSIs yields an indistinguishability of 0.985(1).⁹⁵

2. Reversed Hong-Ou-Mandel

While with the JSI overlap method the two sources are operated independently and only later compared, in the reversed HOM

experiment¹⁰² the two sources are coherently operated, i.e., pumped at the same time with the same pump laser.¹⁰³ The method is based on reversing a HOM experiment, where photon bunching is achieved. Therefore, in a time-reversed HOM, the interference of two indistinguishable photon pairs results in antibunching in the output ports of the beam splitter, i.e., a suppression of two-photon interference. In the experiment schematized in Fig. 9(a), each source generates a signal-idler pair. Through waveguide crossings, the two signal photons and the two idler photons are injected into two different MZIs. The MZIs are operated as a 50/50 beam splitter as in HOM experiments, and the

coincidences are recorded as a function of the phase $\phi_1 = \phi_2 = \phi$ [Fig. 9(c)]. It is possible to demonstrate that the visibility V_{rH} of the observed reversed HOM coincidence fringes is given by the normalized overlap between the two JSAs of the two photon sources,⁹⁵

$$V_{rH} = N^2 \left| \int d\omega_s d\omega_i F_2^*(\omega_s, \omega_i) F_1(\omega_s, \omega_i) \right|, \quad (66)$$

where N is a normalization constant given by

$$N^2 = \frac{2}{\int d\omega_s d\omega_i |F_1(\omega_s, \omega_i)|^2 + \int d\omega_s d\omega_i |F_2(\omega_s, \omega_i)|^2}. \quad (67)$$

Therefore, the visibility V_{rH} is directly proportional to the JSA overlap, i.e., it gives a measure of the indistinguishability $I = V_{rH}$ between the two sources. Note that the reversed HOM visibility does not depend on the separability of the JSA, i.e., it characterizes the source indistinguishability independently from the heralded photon purities.

3. Quantum state tomography

By coherently pumping the two sources, a two-qubit entangled state can be generated. By testing such an entanglement, it is possible to estimate the indistinguishability.⁴¹ Since most of the information of a quantum state is encoded in its density matrix, quantum state tomography is used to estimate the density matrix of the state by a series of measurements.¹⁰¹ Therefore, the setup of Fig. 9(a) is tuned to extract the density matrix of the two-qubit entangled state. This is shown in Fig. 9(d). The fidelity (F_{max}) between the experimental density matrix and the maximally entangled state $|\Phi_+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ can be extracted from these measurements [in the shown example, it is 0.989(3)]. From F_{max} the indistinguishability I can be estimated by $I = 2F_{max} - \rho_{00} - \rho_{11}$, where ρ_{00} and ρ_{11} are the density matrix elements when both the idler (signal) photons are in the top (bottom) output waveguides [in the shown example, $I = 0.982(6)$].

H. Quantum interference

Quantum information and computing require high visibility quantum interference among indistinguishable single photons generated by many different and nominally identical heralded single photon sources. To yield high quantum interference visibility, such photons should be in a pure state indistinguishable, and highly uncorrelated from their twin counterpart. Indeed, high visibility of the quantum interference process is the result of the combined action of fidelity, purity, and indistinguishability.¹⁰⁴ Also, low multipair generation is required, in order to avoid detrimental noise contribution from higher order photon number terms.¹⁰¹ The suitable experiment to on-chip evaluate the quality of a quantum interference is the four-photon heralded HOM experiment,¹⁰⁵ sketched in Fig. 10(a).⁹⁵ Here, one photon in the pair (the herald photon, i.e., the signal photon in this example) is directly detected while the other (the heralded photon, i.e., the idler photon in this example) is used to perform a HOM experiment. Indeed, the two heralded photons are injected in a MZI, where a phase shifter is present on one arm. The relative photon phase θ is scanned, and the interference is recorded by detecting the fourfold coincidences.

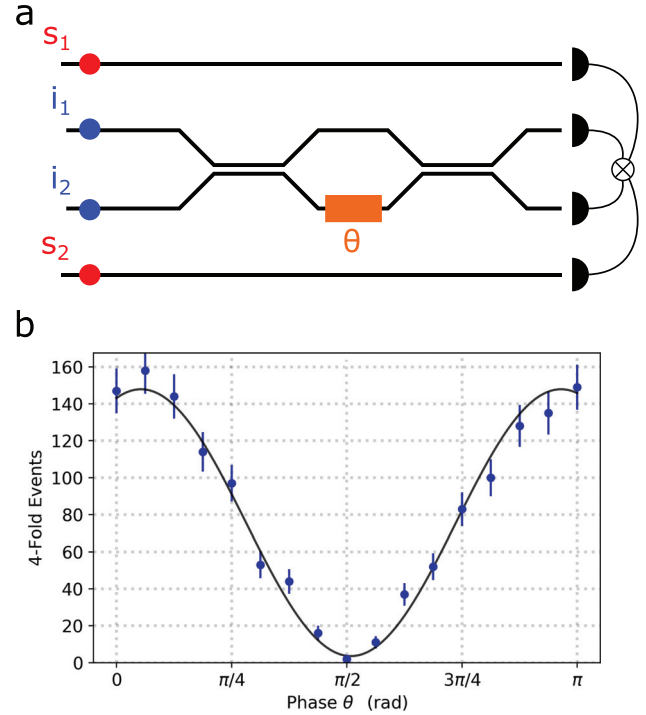


FIG. 10. (a) Experimental setup for the integrated heralded HOM. In the off-chip version, a bulk MZI has to be used. Coincidence measurements from the four detectors are recorded. The phase θ is scanned to measure the dip in the HOM interference. (b) The measured fourfold coincidences of two heralded sources. The measurement exhibits the typical dip of the HOM interference. The measured visibility is evaluated by fitting the data with a sine function. The estimated visibility is 0.96(2). Reprinted with permission from Paesani *et al.*, Nat. Commun. 11, 1 (2020) (Ref. 95). Copyright 2020 Authors, licensed under a Creative Commons Attribution 4.0 International License.

A fringe due to the quantum interference is reported in Fig. 10(b). From these data, the heralded HOM visibility, also known as the contrast, is deduced by

$$V_{HH} = \frac{R_{4s}^{max} - R_{4s}^{min}}{R_{4s}^{max} + R_{4s}^{min}}, \quad (68)$$

where R_{4s}^{max} and R_{4s}^{min} are the measured fourfold coincidence rates at the maximum (split-states) and minimum (bunching) values of the interference fringes. We use the term bunching or split-states because the phase in the MZI allows setting it as a 50:50 beam splitter (i.e., $\theta = \pi/2$, single photons are bunched in one of the two ports) or as a mirror (i.e., $\theta = 0, \pi$, single photons are split and sent to the two detectors). Thus, V_{HH} quantifies the quality of the quantum interference accessible with a heralded single photon source. A value of V_{HH} near one is a critical figure of merit for any heralded single photon source. V_{HH} determines the number of single photon sources in the PIC, i.e., the scalability and complexity of the quantum circuits.¹⁰⁶

IV. STATE-OF-THE-ART

In this section, we will review the state-of-the-art of integrated heralded single photon sources. These can be developed based on

waveguides or microresonators. They differ for the material platforms, for the nonlinear process, for the phase matching geometry, and for the spectral regions. Table II summarizes the representative demonstrations for the different categories.

A. Optical waveguides

Depending on the actual core material, various parametric processes can be excited in a waveguide with different phase matching techniques. For SFWM, phase matching is generally achieved through the engineering of the group velocity dispersion³⁶ or via modal phase matching.⁶¹ For SPDC, modal phase matching or periodically poling technology is used. Since silicon has a strong $\chi^{(3)}$ and a negligible $\chi^{(2)}$ due to its centro-symmetry, SFWM is the standard nonlinear process for silicon photonics. This material platform is the topic of this section. SPDC is only possible in $\chi^{(2)}$ materials and will be analyzed in detail in Sec. IV C. Note that an effective $\chi^{(2)}$ has been induced in silicon by strain^{57,110} or by an external electric field,^{58,111} but SPDC has not yet been observed.

Silicon waveguides can take advantage of the mature CMOS technology, which assures high manufacturing reliability and high process resolution. This is a main advantage of the use of waveguides as heralded single photon sources. Their fabrication does not involve critical design elements (like the gap in microring resonators), which positively affects phase matching requirements and source performances. On the other hand, waveguides have lower brightness and larger footprints than microring resonators.¹⁸

Since the brightness is related to the waveguide length,⁵⁰ spiral waveguides are often employed with the advantage that the spiral geometry minimizes the footprint. Lengths ranging from millimeters to centimeters, with footprints below $250 \times 250 \mu\text{m}^2$, can be achieved.¹⁸ Since waveguides are broadband devices, narrow pass-band filters are needed to select idler and signal photons or to factorize the generated twin photons, so that high purity is possible. Such integrated pass-band filters are centered few nanometers off with respect to the pump wavelength and need a very high rejection rate (90 dB) and a low insertion loss (max 1 dB). In this way, factorizable single photon states are possible at the expenses of the heralding efficiency and of the brightness.

Interesting results have been reported in the silicon-on-insulator (SOI) platform^{43,45} and in the silica-on-silicon platform.^{86,107} In particular, four heralded single photon sources based on silicon spiraled waveguides were coupled with reconfigurable quantum gates and qubit projectors in a single chip [Fig. 11(a)].⁴⁵ This SOI chip implements four photon graph states. The single photon sources have a purity between 82% and 92% and an intrinsic heralding efficiency of 60% despite the used off-the-chip narrow (0.7 nm) bandpass filters. The overall high quality of the sources is measured through a heralded HOM experiment, which yields a visibility of 82(2)%.

High quality heralded single photon sources have been achieved with silica-on-silicon waveguides.¹⁰⁷ An array of five straight waveguide sources has been fabricated by direct UV laser writing in silica. The waveguides have circular cross sections [Fig. 11(b)]. The circular shape allows minimizing the coupling losses with optical fibers. In addition, SFWM is achieved via birefringent phase matching,⁵⁵ which yields discrete band phase matching and limited spectral correlations. Thus, only moderate spectral filtering (5 nm and 20 nm band-pass filters for the signal and idler, respectively) is required as shown in the simulated JSI [Fig. 11(b)]. These silica waveguide sources show $\eta_I \geq 66\%$ ($\eta_F > 92\%$), with a theoretical $P = 97\%$, $I = 98.0(4)\%$, and a raw $V_{HH} = 90(2)\%$. With respect to silicon, silica shows lower propagation losses (3 dB/cm in Si vs 0.4 dB/cm in silica) and birefringent phase matching, but also lower refractive index contrast (i.e., larger footprint) and lower nonlinearity, about one thousand times weaker.¹¹² It is difficult to foresee a large scale integration of quantum photonic circuits in silica.

A fundamental trade-off between purity, on one side, and heralding efficiency and brightness, on the other side, affects waveguide sources due to the need of the band-pass filters.^{96,113,114} Therefore, decoupling purity and heralding efficiency is crucial to near-ideal heralded single photon sources. This was achieved by the combination of intermodal SFWM⁶¹ and delayed pump excitation^{88,89,115} in silicon spiral waveguides.⁹⁵ The PIC design is shown in Fig. 11(c), where two pump pulses are propagating on two different modes with different group velocities in a multimode waveguide (propagation losses < 0.4 dB/cm, thanks to the large waveguide width). An on-chip delay line makes the two pump pulses to overlap only when both are completely loaded in the multimode waveguide. As a result, the idler and signal photons are generated in a factorizable state with

TABLE II. A selection of representative heralded single photon sources. The symbols in the column headings refer to the ones used in Table I.

Material	Process	Wavelength (nm)	η_I (%)	I (%)	P (%)	V_{HH} (%)	References
Optical waveguides							
Silicon (spiral waveguide)	SFWM	1550	60	...	92	82 (2)	45
Silica on silicon (straight waveguide)	SFWM (birefringent PM)	815–820	66	98.0 (4)	97	90 (2)	107
Silicon (spiral waveguide)	SFWM (intermodal and delayed pump)	1550	91 (9)	98.7 (2)	99.04 (6)	96 (2)	95
Optical microresonators							
Silicon (dual-MZI-coupled ring)	SFWM	1550	52.4	...	95 (1.5)	...	108
Silicon (point-coupled ring)	SFWM	1550	...	90.99 (3.91)	84	87.3 (2)	44, 104
AlN (point-coupled ring)	SPDC	1550	...	...	107 (12)	...	76
Periodically poled waveguides							
Ti:LiNbO ₃	Type II SPDC	1552	63	...	93.5 (2)	...	109

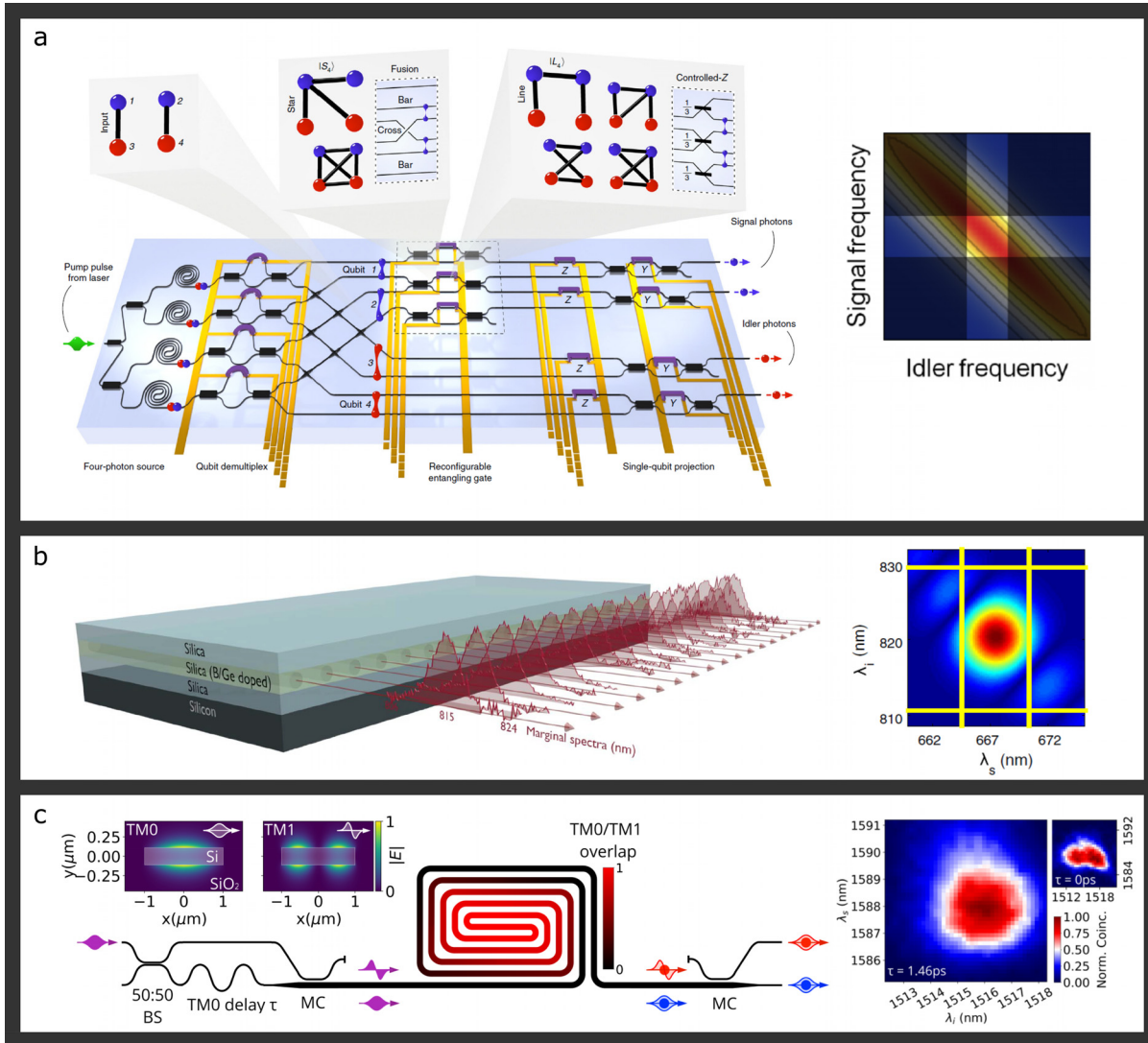


FIG. 11. (a) (Left) Four heralded single photon sources based on silicon spiraled waveguides, coupled with reconfigurable quantum gates and qubit projectors on the same chip. The top images indicate the generated graph states. (Right) JSI of the sources after on-chip filtering. Reprinted with permission from Adcock *et al.*, Nat. Commun. **10**, 1 (2019) (Ref. 45). Copyright 2019 Authors, licensed under a Creative Commons Attribution 4.0 International License. (b) (Left) Array of heralded single-photon sources on a silica photonic chip. By SFWM, idler and signal photons are generated at 820 nm and 667 nm, respectively. (Right) Simulated JSI of the source. The yellow lines indicate the bandpass filters used to improve the purity of the source. Reprinted with permission from Spring *et al.*, Optica **4**, 90 (2017) (Ref. 107). Copyright 2017 Authors, licensed under a Creative Commons Attribution 4.0 International License. (c) (Left) Schematics of the intermodal heralded single photon source. The TM0 mode is delayed with respect to the TM1 pulse. The inset shows the spatial profile of the TM0 and TM1 modes. The maximum overlap between the two pulses is achieved in the middle of the spiral waveguide, where there is the red colored region. Asymmetric directional couplers are used as mode converters (MCs). (Right) Measured JSIs with and without a delay of $\tau = 1.46$ ps. Reprinted with permission from Paesani *et al.*, Nat. Commun. **11**, 1 (2020) (Ref. 95). Copyright 2020 Authors, licensed under a Creative Commons Attribution 4.0 International License.

$P = 99.04(6)\%$ [see the circular JSI in Fig. 11(c)]. This high factorizability comes from the absence of the sinc lobes in the generated spectra of the signal and idler, which are now Gaussian shaped thanks to the matched delay between the pumps.¹¹⁶ Intermodal phase matching allows generating the single photons in discrete bands, therefore, broad band filters are used only to remove the residual pump. In this way, $\eta_F \geq 99\%$ and $\eta_I = 91(9)\%$, $I = 98.7(2)\%$ and $V_{HH} = 96(2)\%$.

These numbers place this source among the best ever realized in a PIC.

A final note is about the parasitic nonlinear effects which mainly affect silicon waveguides. Due to the high third-order nonlinearity and the small mode size, silicon waveguides have a sizable two-photon absorption (TPA), which limits the heralding efficiency¹¹⁷ and leads to the generation of a large concentration of free carriers.¹¹⁸ These in

turn affect the silicon optical properties by free carrier dispersion and free carrier absorption.¹¹⁹ Free carrier effects can be mitigated by using a lateral p-i-n junction across the waveguide to deplete the core of the waveguide from free carriers.¹²⁰ Since TPA is an intrinsic silicon property, it imposes a strong limit to the source performance by causing loss to the pump (TPA) and to the generated photons (cross-TPA).¹¹⁸ Modeling of the effects of TPA on single photon generation has been also extended to silicon microresonators.¹²¹

B. Optical microresonators

Optical microresonators enhance the intra-cavity power, which triggers parametric processes with higher efficiency than in waveguides.^{17,36} Microdisk^{94,122} and microring^{76,108} resonators are the most used geometries in PIC due to the ease of design and fabrication. The microresonators should be doubly (SPDC) or triply (SFWM) resonant, in order to efficiently couple pump, signal, and idler modes.¹²³ Phase matching is not a concern because the cavity resonances are equally spaced in frequency. Moreover, for SFWM, being signal and idler symmetric with respect to the pump, evenly spaced resonant frequencies are required. These are usually neighboring resonances, where the dispersion is linear.¹⁸ Resonator based heralded single photon sources easily surpass their waveguide counterparts in terms of power efficiency and brightness,⁵⁰ CAR (hundreds for waveguides^{124,125} and above thousands for resonators^{122,126}), and footprints [$\leq 50 \times 50 \mu\text{m}^2$ for a microring radius between $5 \mu\text{m}$ and $30 \mu\text{m}$ (Ref. 127)]. A limitation is the strong dependence of the resonance wavelengths on the temperature and on the fabrication accuracy, which reduces the brightness and the indistinguishability of the source. A mitigation is to use an active control, which is typically realized by integrated thermal resistors.

Let us concentrate on microring resonators side-coupled to bus-waveguides. A sketch of this device is reported in Fig. 12. Its performances are determined by the quality factor Q , which relates the resonator circulating power to the resonator losses. The Q -factor is linked to two coupling efficiencies: K_c , between the bus-waveguide and the resonator, and K_s , between the resonating mode and the scattering modes (related to the resonator losses). The Q -factor directly affects the pair generation rate (PGR) of the SFWM process which is maximized at the critical coupling ($K_c = K_s$).¹²⁸ However, this condition limits the

heralding efficiency since the heralded photons can be lost either in the scattering mode or in the waveguide mode with a 50% probability.¹²⁹ Thus, a trade-off between the brightness and the heralding efficiency exists. In fact, the heralded photon flux J_h can be modeled as¹²⁹

$$J_h \propto \frac{K_c^4}{(K_c + K_s)^6}, \quad (69)$$

while η_I is given by

$$\eta_I = \frac{K_c}{K_c + K_s}. \quad (70)$$

At the critical coupling $\eta_I = 0.5$ and $J_h \propto K_c^{-2}$, while in the overcoupling regime ($K_c \gg K_s$) $\eta_I \rightarrow 1$ and $J_h \rightarrow 0$ or in the undercoupling regime ($K_c \ll K_s$) $\eta_I \rightarrow 0$ and $J_h \rightarrow 0$. Therefore, a balance between resonator losses and microring/waveguide coupling should be looked for. A crucial role is played by the resonator losses, which set the maximum J_h for a fixed η_I . This trade-off also impacts the purity of the heralded state. Indeed, for pair generation in a point-coupled bus-waveguide/microring resonator system, the biphoton wavefunction, in Eq. (40), can be written as⁹⁰

$$F(\omega_s, \omega_i) \propto F_p(\omega_s + \omega_i) l_s(\omega_s) l_i(\omega_i), \quad (71)$$

with

$$F_p(\omega) = \int d\omega_p \alpha_p(\omega_p) \alpha_p(\omega - \omega_p) l_p(\omega_p) l_p(\omega - \omega_p), \quad (72)$$

and α_p being the pump spectral profile, $|l_p|^2$, $|l_s|^2$, $|l_i|^2$ being the Lorentzian lineshape functions of the pump, signal, and idler resonances. To generate pure heralded single photons, $F(\omega_s, \omega_i)$ should be separable, i.e., $F_p(\omega_s + \omega_i)$ should be constant within the resonances. This practically translates to the need of a pump that is spectrally broader than l_p . Increasing the pump bandwidth even more does not result in an improved separability unless the resonance linewidth also increases, which in turn means a lower Q -factor. In a resonator where all the resonances involved have the same linewidth, the fundamental upper limit to the purity is $P = 93\%$.⁹⁰ Therefore, purity requires a broad pump resonance, i.e., the microring should be set in an overcoupling regime where J_h decreases.

At this point, it is interesting to compare the emission spectrum of waveguide based sources with the one of microring based source. Considering SFWM, the emission spectrum of a waveguide is typically a broad band centered at the pump wavelength, as shown in Fig. 13(a). To select the idler and the signal wavelengths, filtering is required, which results in high losses and limited brightness. Filters are no longer needed with intermodal SFWM, whose generation spectrum [see Fig. 13(b)] exhibits discrete bands. This is similar to the microring generation spectrum shown in Fig. 13(c). Both the intermodal as well as the microring approaches do not require tight filtering to further select the emission modes. This increases the heralding efficiency. However, the purity of heralded single photons generated by microring resonators is limited by the linewidths of the microring resonances. Indeed, the modal structure of the microring affects the phase matching bands of SFWM. Moreover, increasing the pump bandwidth does not result in improved purity, being limited by the resonance linewidth at the pump frequencies. Microring resonators exhibit high purity and low losses. At present, they cannot achieve the purity and heralding

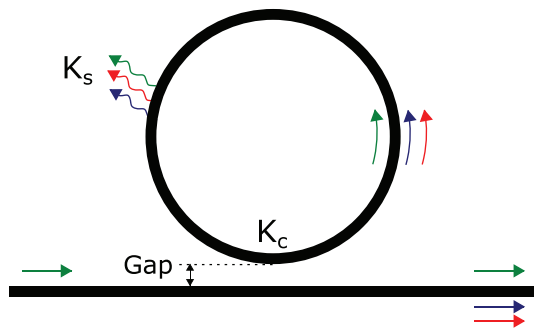


Fig. 12. Schematics of a microring resonator side-coupled to a bus-waveguide. The green arrow refers to the pump; the blue and red arrows refer to the generated idler and signal photons. K_s is the coupling coefficient between the resonating modes and the scattering modes. K_c is the coupling coefficient between the bus-waveguide and the microring, which depends on the gap between them.

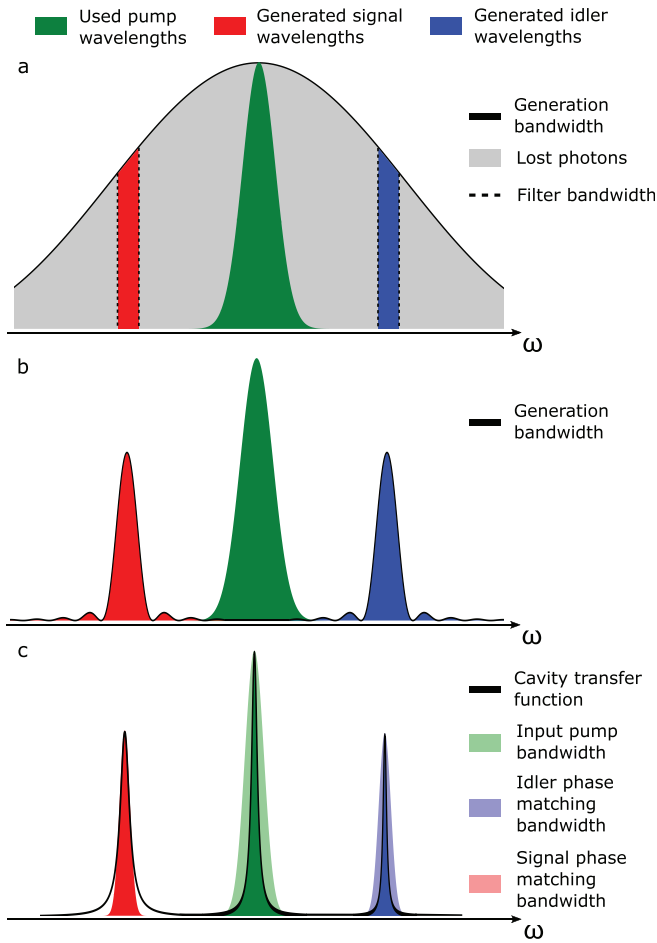


FIG. 13. Emission bandwidths for intramodal SFWM in a waveguide (a), for intermodal SFWM in a waveguide (b), and for SFWM in a microring resonator (c). The dark line refers to the SFWM generation bandwidth which is broad for the waveguide (a), has a discrete narrow band for intermodal phase matching (b), and is determined by the modal structure in a microring resonator (c). The green narrow band refers to the pump spectrum. The red and the blue bands indicate the generated spectral distribution of signal and idler photons. For intramodal SFWM (a), the idler and signal photons are spectrally filtered to purify the resulting two-photon state. In the case of intermodal SFWM (b), the idler and signal photons are windowed by the phase-matching bands. As a result, filters are not required to get high purity and the heralding efficiency can be made close to unity. The emission of SFWM in a microring cavity (c) is filtered by the resonator modes.

efficiency of intermodal SFWM waveguides, which in turn are characterized by a brightness lower than microring resonators. In fact, a pair generation efficiency of about 20×10^6 counts/s/mW² is demonstrated in waveguides⁹⁵ to be compared with the microring results described in Ref. 44 (pair generation efficiency of 55×10^6 counts/s/mW²). Thus, waveguide sources are more demanding in terms of input power, but they show brightnesses comparable with microrings. On the other hand, microring based sources cannot offer the same level of purity, indistinguishability, and heralding efficiency of waveguides when intermodal SFWM is used.

To overcome the limitations set by the purity–brightness trade-off, the signal, idler, and pump resonance lineshapes have to be controlled independently. Two main approaches have been proposed. The first one is based on the interference between two delayed pump pulses.⁸⁸ By tuning the pulse spectrum and delay, destructive interference in the microring and constructive interference in the bus-waveguide are achieved. As a result, the pump light is extracted from the resonator with a much shorter time than the standard single pulse pumping. This broadens the pump resonance, while not affecting the idler and signal resonances. Theoretically, an enhanced purity up to 0.999 is predicted,⁸⁸ as shown in Fig. 14(a). A second strategy is based on the individual tuning of the resonances Q-factors with a dual-MZI coupled silicon microring geometry,^{90,108} sketched in Fig. 14(b). The transfer function of the two asymmetric MZI and the resonances of the microring are controlled by thermo-optic phase shifters to tune independently their Q-factors and spectral position.^{130,131} Optimizing the device without bandpass filters, $P = 95(1.5)\%$, $\eta_I = 52.4\%$, and a CAR of 81 with a measured coincidence of 9599 pairs/s [pair generation rate (PGR) of 1147k counts/s] were demonstrated.¹⁰⁸ The purity now directly depends on the ratio between the signal/idler quality factor and pump quality factor, as measured in Fig. 14(b).

If the dual-MZI-coupled microring resonator demonstrated the highest purity, the point-coupling microring resonator was used to demonstrate quantum interference and indistinguishability.^{44,76,98,104,126,132} Four identical all-pass microring heralded single photon sources were fabricated on a chip [Fig. 14(c)] and coherently pumped.⁴⁴ Phase shifters allow tuning the different microring resonances to maximize their indistinguishability up to $I = 90.99(3.91)\%$ and raw $P = 84\%$, with CAR = 350 and $g_h^{(2)} < 0.01$ at about 3k counts/s coincidence and raw $V_{HH} = 71.9(2.4)\%$ [87.3(1.9)% with multipair correction], as shown in Fig. 14(c). With these heralded sources, on-chip genuine multipartite entanglement and quantum teleportation have been demonstrated, confirming the suitability of microring resonators for complex quantum networks.

On-chip free standing silicon microdisk resonators have been also fabricated with $P \geq 90\%$, $\eta_I \approx 50\%$ and CARs ranging from hundreds to tens of thousands.^{94,122} An illustration of the device and the corresponding CAR and purity measurements are reported in Fig. 14(d). With respect to microring resonators, free standing microdisk resonators are complicated by the tapered fiber coupling, which is unstable and does not allow easy integration.

Second order nonlinear semiconducting materials have been employed to fabricate heralded single photon sources in PIC by using microring resonators. The best results have been reported by using aluminum nitride (AlN) on silica.⁷⁶ AlN is a $\chi^{(2)}$ material suitable for low footprint and scalable photonics.¹³³ By SPDC in a microring resonator source, sketched in Fig. 14(e), twin photons were generated at 1550 nm by pumping at 775 nm which required a careful design due to the very different wavelength ranges. This difference, on one side, can be used to efficiently filter the pump photons and, on the other side, requires a multimodal phase matching, where the pump photon and the generated photons are coupled to different modal orders. $P = 1.07(12)$, $g_h^{(2)} = 0.088(4)$ [Fig. 14(e)] at a 1.9 mW on-chip CW power, PGR = 11.0 M counts/s, and 80 counts/s coincidence rate were demonstrated.⁷⁶

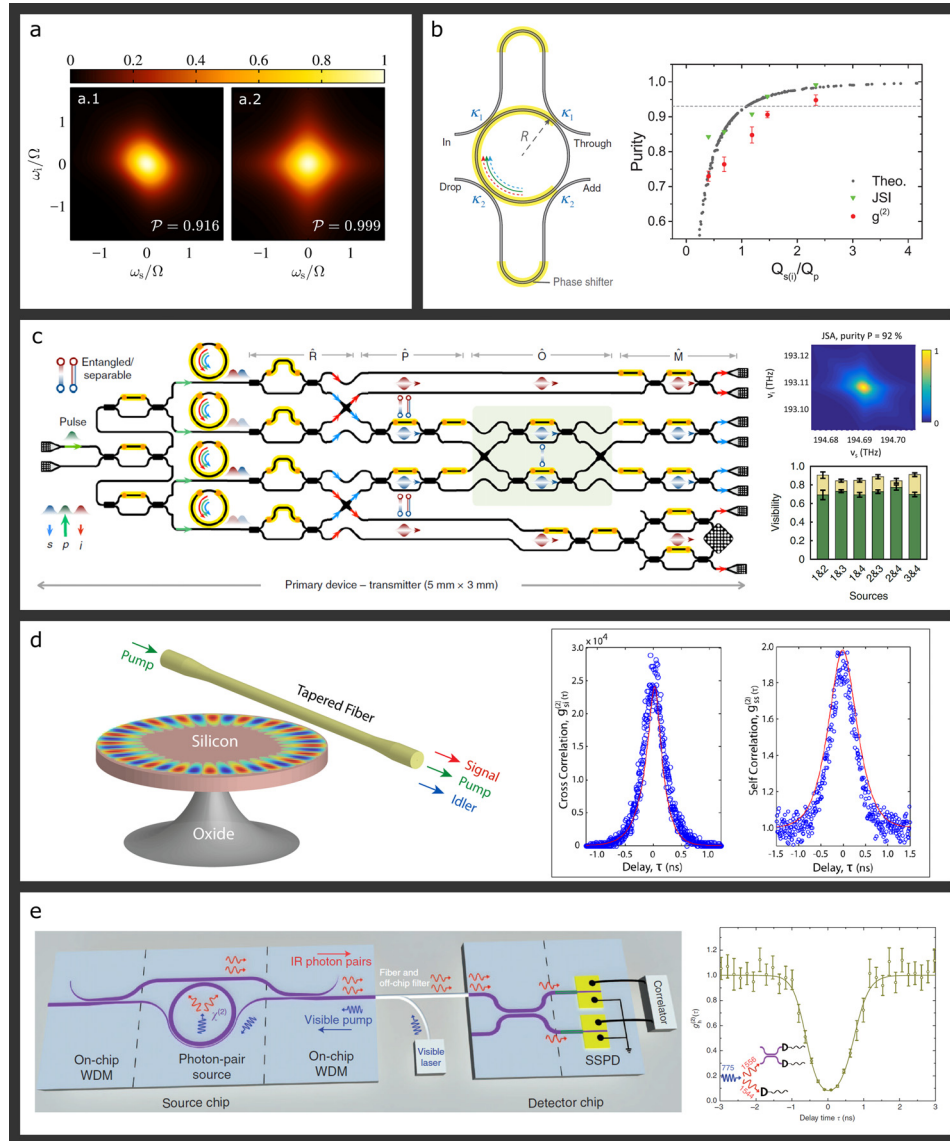


FIG. 14. (a) Joint spectral intensity calculated for a microring resonator, where the delayed pump technique has been applied. (a.1) and (a.2) refer to the single-pulse and the dual-pulse cases, respectively. The purity P is reported in the inset. Axes are normalized with respect to the resonance frequency. Reprinted with permission from Christensen *et al.*, *Opt. Lett.* **43**, 859 (2018) (Ref. 88). Copyright 2018 Optical Society of America. (b) On the left, a sketch of the dual-asymmetric MZI coupled microring. Two different point coupling efficiencies (κ_1) and (κ_2) are used. Phase shifters (in yellow) allow tuning the phase in the asymmetric MZI as well as the microring resonance. On the right, the purity as a function of the ratio between the signal (idler) quality factor and the pump one. The calculated values (Theo.), measured values via stimulated emission tomography (JSI), and $g^{(2)}$ analysis. Reprinted with permission from Liu *et al.*, *Opt. Lett.* **45**, 73 (2020) (Ref. 108). Copyright 2020 Optical Society of America. (c) Schematics of a silicon chip with four identical microring resonator SFWM sources. Blue and red arrows refer to signal and idler photons, respectively. The chip can be reconfigured to prepare the photons in entangled or separable states. The generated photons are demultiplexed through MZIs and routed by using waveguide crossings (R). After being prepared (P), photons go through a linear-optic quantum circuit (O) and are subjected to the projective measurements (M). Yellow dots label the thermo-optic phase shifters. The top inset shows the simulated JSI with an expected purity of 92%. The bottom inset shows the measured quantum interference visibility between two microring sources. The values are reported with (green) and without (yellow) multipair corrections. Reprinted with permission from Llewellyn *et al.*, *Nat. Phys.* **16**, 148–153 (2020) (Ref. 44). Copyright 2020 Springer Nature. (d) Schematics of a free standing silicon microdisk resonator on a silica pedestal coupled to a tapered fiber. Inset: cross-correlation ($g_{si}^{(2)}$) between the signal and idler photons, showing a peak value of $2.58(16) \times 10^4$, and signal self-correlation ($g_{ss}^{(2)}$), demonstrating a purity of 0.90 (5). Reprinted with permission from Lu *et al.*, *ACS Photonics* **3**, 1626 (2016) (Ref. 122). Copyright 2016 American Chemical Society. (e) (Left) Schematics of the AIN microring resonator source which generates twin-photons via SPDC. On-chip wavelength division multiplexers (WDM) have been used to outcouple selectively the generated pairs. The generated pairs are then coupled to another chip with an integrated beam splitter and superconducting single photon detectors (SSPD). (Right) Measured $g^{(2)}(0)$ showing the antibunching dip of the heralded single photon source statistics. The inset reports the experimental setup. Reprinted with permission from Guo *et al.*, *Light* **6**, e16249 (2017) (Ref. 76). Copyright 2017 Authors, licensed under a Creative Commons Attribution 4.0 International License.

C. Periodically poled waveguides

The most valuable improvements in terms of complexity and integration for chip scale quantum networks have been driven by silicon photonics, with waveguides and microring resonators as the fundamental building blocks of the quantum circuit.^{14,41} However, an alternative platform for heralded single photon sources is based on Lithium Niobate (LiNbO₃, LN for short)¹³⁴ or potassium titanyl phosphate (KTiOPO₄, KTP for short).¹³⁵ These are $\chi^{(2)}$ materials where periodically poled waveguides are formed to enable quasi-phase-matched SPDC [Fig. 15(a)].⁹⁹ Sources based on $\chi^{(2)}$ materials can profit from the high efficiency of SPDC, the unique fast (GHz) electro-optic effect, and the large detuning between the pump and the generated twin photons.¹³⁶ These benefits are counteracted by other limitations associated with the cost, the large footprint in the cm² range, the need of temperature trimming of the phase matched wavelength, and the complex not-CMOS fabrication processes.

To achieve separable states from SPDC, the group velocity matching (GVM) technique is used [Fig. 15(b)]. In fact, the biphoton wavefunction in Eq. (40) is the product between the pump spectrum and the phase matching function $\phi(\omega_s, \omega_i, \omega_p)$. $\phi(\omega_s, \omega_i, \omega_p)$ is determined by the relationship between the pump, signal, and idler group velocities.¹³⁷ Matching properly the group velocities yields an improved separability of the biphoton wavefunction, which enables heralded single photon sources with high purities without the use of spectral filtering.^{75,93,99,109,137}

Figure 16(a) shows an example of a PIC with a heralded single photon source, an electro-optical router, and a voltage controllable time delay line integrated in Ti:LiNbO₃.¹⁰⁹ This is almost the only example of a PIC integrating a tunable delay line. The source is a periodically poled waveguide pumped at 776 nm and generating degenerate twin photons at 1552 nm. By applying a proper polarization rotation and time delay, on-chip HOM measurements [Fig. 16(a)] between the two degenerate orthogonal polarized photons yield $P = 93.5(1.8)\%$ with a $\eta_I = 63\%$, thanks to the low propagation (0.1 dB cm⁻¹) and coupling losses. Also, hybrid solutions have been

proposed where periodically poled LN waveguide sources are coupled with glass based laser written PIC.¹³⁸ In this way, the high generation efficiency of LN is combined with the flexible and low loss glass platform. An on-chip source of reconfigurable heralded entangled states (N00N and separable states) was achieved by thermal tuning the quantum circuit, whose schematics has been reported in Fig. 16(b).

D. Multiplexed heralded sources

Applications like quantum computing and communication require a high number of interfering single photons. To make these interferences likely to occur, deterministic single photon sources are needed. This is the main limit of heralded single photon sources: they cannot guarantee a high probability of single photon emission and, as a consequence, of simultaneous emission and interference from separate sources.⁴⁸ In fact, due to the use of nonlinear spontaneous processes, the maximum single-photon generation probability of a heralded single photon source is limited to 25%.¹¹³ This is related to the thermal emission statistics of a single mode parametric source, where $P(1)$ is related to $P(n > 1)$ via μ . This dependence is the reason for the low brightness of heralded single photon sources, since a low μ is required to maximize $P(1)/P(n > 1)$.⁹⁷ A solution to this problem is to multiplex an array of heralded single photon sources, so that the aggregated $P(1)$ increases without increasing $P(n > 1)$. This approach relies on the detection of the herald photon to route deterministically the corresponding heralded photon to the output of the multiplexed source. In this way, several probabilistic sources are combined into a near-deterministic multiplexed one. In more practical terms, a multiplexed source keeps the same $g^{(2)}(0)$ of the individual sources with an enhanced emission rate.

Multiplexing can be based on spatial, temporal, or frequency degrees of freedom. When pair emissions occur in spatially separated modes, these can be combined into a common output mode.³⁷ Several demonstrations of spatial multiplexing have been reported on PIC.^{38,139–141} As an example, two nominally identical photonic crystals waveguides (PhCWs) were multiplexed on a silicon chip [see Fig. 17(a)].³⁸ $R_h = 5$ counts/s (increased by 63.1% with respect to a single PhCW) with $g^{(2)}(0) = 0.19(2)$ was reported. Further improvements can be obtained by a bidirectional pump configuration yielding a further increase in R_h by 51(4)%, while maintaining the same single photon probability emission.¹⁴⁰ Combining the bidirectional pumping with spatial multiplexing provides a twofold increase in the final footprint and number of multiplexed sources. Despite the straightforward implementation, spatial multiplexing becomes quickly not practical when the number of multiplexed sources scales up. In fact, not only does the footprint increase almost linearly with the number of multiplexed sources, but also the number of detectors, switches, and delay lines.¹⁴² In fact, multiplexing requires switches to route the generated photons to the common output, detectors to detect the herald photons, and delay lines to store the photons or to match the optical with the electronic timing. To accommodate all these components, the PIC gets larger in size which increases the propagation losses due to the increased optical path.¹⁴³

Alternatively, temporal multiplexing offers most of the advantages of spatial multiplexing, reducing the number of optical components in the chip.⁹⁷ This scheme involves the generation of twin photons in successive temporal modes, which are multiplexed to give a large R_h . If the pump has a given repetition rate R_p , by multiplexing

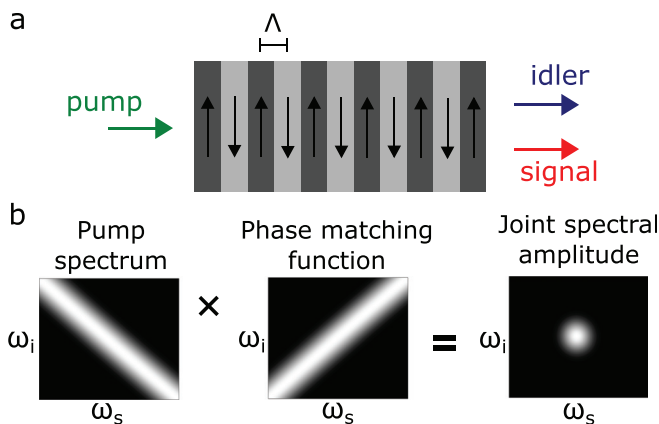


FIG. 15. (a) Sketch of a periodically poled waveguide. Λ is the poling period. (b) Schematics of the GVD technique to achieve a factorizable JSA. By engineering the GVDs of the poled waveguide, the phase matching function can be rotated in the (ω_s, ω_i) plane so that its product with the pump spectrum gives a circular, i.e., factorizable, JSA.

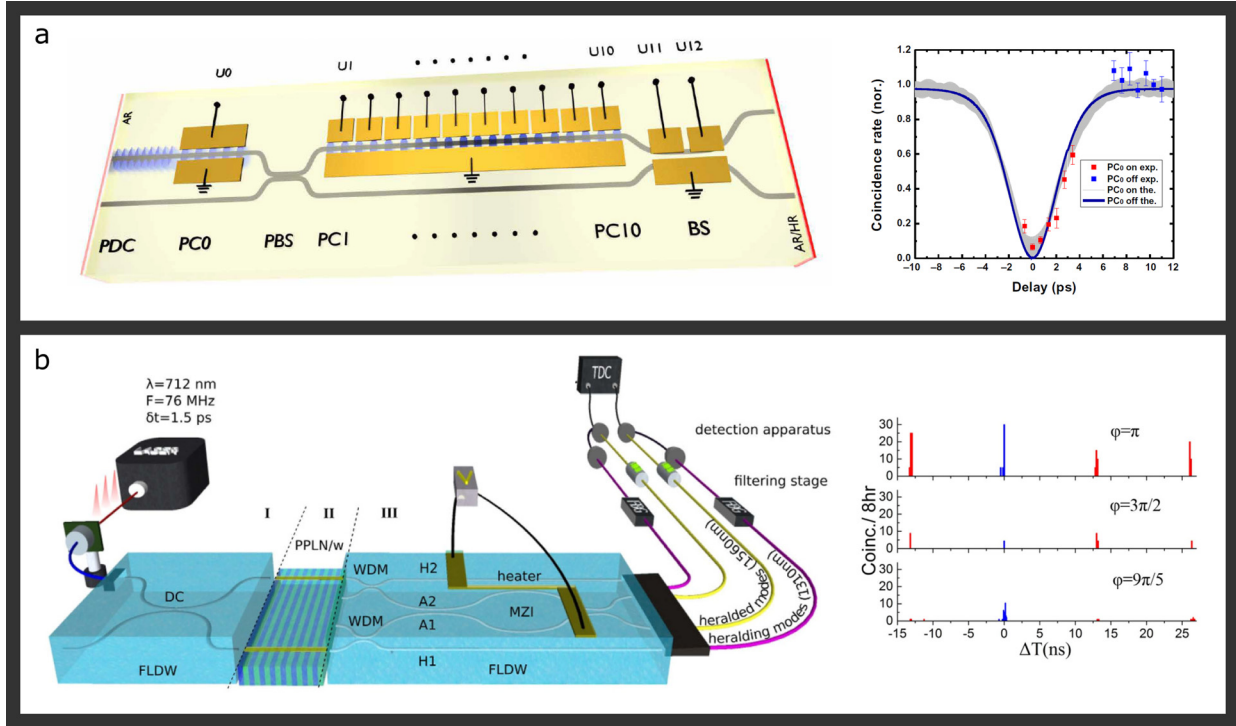


Fig. 16. (a) (Left) Schematics of the Ti:LiNbO₃ PIC. It integrates a periodically poled PDC twin-photon source, an electro-optical router, and a voltage controllable delay line. These are made using a polarized beam splitter (PBS), a 50/50 beam splitter (BS), and polarization controllers (PC0–PC10) driven by voltages (U₀–U₁₂). Antireflection (AR) and high reflection (HR) coatings are used on the crystal facets. (Right) Measured (exp.) and simulated (theo.) HOM interference measured by delaying the signal and idler photons with the PC0. Reprinted with permission from Luo *et al.*, *Sci. Adv.* **5**, eaat1451 (2019) (Ref. 100). Copyright 2019 Authors, licensed under a Creative Commons Attribution 4.0 International License. (b) (Left) Schematics of the hybrid periodically poled LN-glass hybrid chip. A pulsed laser centered at 712 nm is used to excite SPDC in two PPLN (Periodically Poled Lithium Niobate) waveguides, which are coupled to the glass PIC. The optical circuit is laser written in silica. The output photons are separated in heralding (violet) and heralded (yellow) photons. The detectors are connected to a time to digital converter (TDC). (Right) Start-stop measurements of the coincidences with different configurations of the chip. Different phases ϕ induced by the heaters lead to different coincidence statistics between the heralded modes. Reprinted with permission from Vergyris *et al.*, *Sci. Rep.* **6**, 35975 (2016) (Ref. 138). Copyright 2016 Authors, licensed under a Creative Commons Attribution 4.0 International License.

n successive pulses, an enhancement in $P(1)$ is obtained. Time multiplexing requires delay lines in order to synchronize the multiplexed pulses. This is its main limitation due to the high insertion losses and limited delays of on-chip delay lines.⁹⁷ In fact, the experimental demonstrations of temporal multiplexed sources reported to date involve integrated heralded single photon sources coupled to off-chip multiplexing architectures. Therefore, both spatial and temporal multiplexing methods have limitations. The first can be fully implemented on chip while the second requires coupling to off-chip macroscopic delay lines. An example of a time multiplexed integrated source is shown in Fig. 17(b).¹⁴⁴ Four pump pulses, spaced by a period T , are delayed and temporally overlapped in a silicon waveguide. All the pulses have the same average power; thus, the SFWM probability is the same for all. However, multiplexing these four independent temporal modes increases linearly the twin photon generation probability. The net increase in R_h is witnessed by a twofold improvement in CAR [Fig. 17(b)], while the high quality of the multiplexed source is given by $V_{HH} = 91(16)\%$.

Frequency multiplexing is a further option. However, to date, it has been implemented only with not integrated heralded single photon sources.^{39,145} Broadband SPDC processes in a periodically poled LN

crystal were used to generate single photons with different frequencies which are then multiplexed in a single optical fiber [Fig. 17(c)].³⁹ The heralded photons are coupled in the same optical fiber together with an array of pump lasers with different frequencies. These pump lasers are switched on by the detection of the herald photons, in order to spectrally translate the heralded photons into a common spectral mode. Therefore, depending on the heralding, a different pump is coupled in the same fiber with the heralded photon, converting via stimulated FWM the heralded photons into a unique output sequence of spectrally homogeneous single photons. In this way, a net enhancement of 3.5 dB in the single photon rate and a doubling of the CAR were observed [Fig. 17(c)]. One of the main advantage of this approach is that the heralded photons do not propagate through multiple switches, considerably lowering the total losses of the system. As a perspective, bulk heralded single photon sources are a beacon for integrated solutions. Indeed, remarkable results have been reported via a large-scale time multiplexed source. Multiplexing 40 temporal generation modes coming from a 20 mm-long periodically poled KTP crystal in an ultralow loss free space storage loop enabled a 9.7(5) dB enhancement with respect to the not multiplexed source, with a maximum single photon probability of 0.667(24).⁴⁰

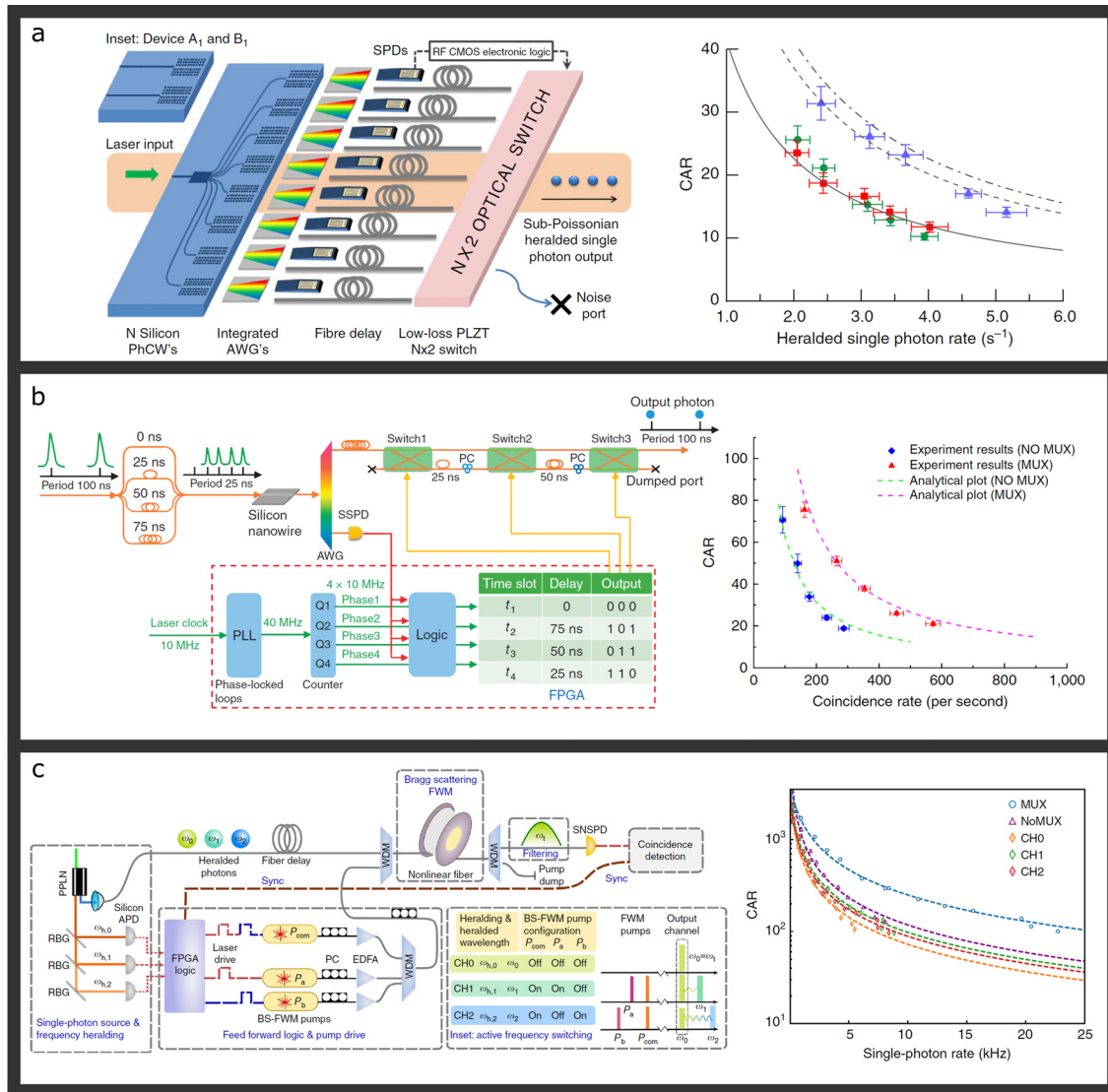


Fig. 17. (a) (Left) Schematics of the spatial multiplexing used in Ref. 38. A pump is coupled to N identical PhCWs to excite SFWM. The idler and signal photons are separated by means of AWGs. Each herald photon is detected via single photon detectors (SPDs) to trigger the routing of the corresponding heralded photon to the output of the electro-optic switches. The inset shows two nominally identical PhCWs (A_1 and B_1). (Right) Measured CAR as a function of the heralded single photon rate for the two individual sources, A_1 (green data) and B_1 (red data), and for the multiplexed source (blue data). The solid black line is a fit of the single source measurements. The dashed line is a fit to the multiplexed results. (b) (Left) Schematics of the active temporal multiplexing scheme used in Ref. 144. The pump pulses are delayed to reduce their time spacing from 100 ns to 25 ns. These pulses are then coupled into a silicon waveguide, where SFWM takes place. An AWG separates the herald from the heralded photons. The heralding signals are detected and sent to a FPGA (Field-Programmable Gate Array), which configures the switches to route the heralded photons into the same output temporal mode. When the logic output is “0,” the photon remains in the input channel; when it is “1,” the photon is routed to the cross channel. In this way, the resulting output mode has again a periodicity of 100 ns, but with enhanced generation probability. (Right) CAR as a function of the coincidence rate for the individual source (NO MUX) and the multiplexed one (MUX). Reprinted with permission from Xiong *et al.*, Nat. Commun. 7, 1 (2016) (Ref. 144). Copyright 2016 Authors, licensed under a Creative Commons Attribution 4.0 International License. (c) A periodically poled LN is pumped to generate twin-photons by SPDC. The herald photons are divided into three different spectral modes and detected by silicon avalanche photodiodes (APD). The heralded photons are coupled to the same single mode fiber. The heralding detectors, after being processed by a FPGA, trigger the selection of the Bragg Scattering-Four Wave Mixing (BS-FWM) pumps, that are coupled together with the heralded signals in the same nonlinear fiber. In the nonlinear fiber, the heralded photons are spectrally converted, by means of BS-FWM, into the same frequency mode, that is filtered and, then, detected via a superconducting nanowire single photon detector (SNSPD). A synchronized coincidence detection is then performed between the heralded and the herald detections. In the inset, the logic used to control the BS-FWM pumps is shown. The heralding modes, belonging to three different spectral channels (CH), trigger the conversion of the heralded photon frequencies to the same target frequency (ω_i). (Right) CAR vs single-photon rate. The multiplexed source (MUX) exceeds the not multiplexed source (NoMUX) by a factor of two. Also, the individual channels (CH 0-2) are reported. Reprinted with permission from Joshi *et al.*, Nat. Commun. 9, 1 (2018) (Ref. 39). Copyright 2018 Authors, licensed under a Creative Commons Attribution 4.0 International License.

Finally, the relative multiplexing technique has been proposed to minimize the requirements for the active switching used in multiplexing.¹⁴² It is based on the recognition that linear-optical quantum computing architectures do not require all (or even most) events to be synchronized. For example, in the two photon quantum interference, only the relative indistinguishability is relevant. Therefore, changing the spatial mode or the time bin of one of the two photons rather than of both is typically sufficient. This relative synchronization reduces the number of switches, decreases the on-chip losses, and improves the multiplexing efficiency.

V. CONCLUSION

Quantum information applications and protocols require a large number of interacting qubits.^{146,147} Integrated heralded single photon sources are the only solution for reliable quantum architectures with many single photon sources. Integrated solutions are also cost effective, less energy consuming, feasible, and reliable in terms of fabrication. Silicon photonics has experienced the most prominent improvement in terms of quantum circuit complexity, with up to 16 pair sources, 61 MZIs, and > 90 heaters on the same chip.⁴¹ The maturity of silicon technology has no equal within the photonics framework, especially in terms of fabrication reliability and footprint. However, silicon photonics for quantum applications mostly relies on third order nonlinear processes (i.e., SFWM sources), less efficient with respect to second order SPDC. This is one of the main limitations of this platform. Because of this, $\chi^{(2)}$ materials are still in the race for the definitive platform for integrated quantum photonics. $\chi^{(2)}$ materials can leverage SPDC, more efficient than SFWM, for the generation of photon pairs and the electro-optics effect for ultra-high frequency modulators on the same chip.^{148,149} Moreover, periodic poling quasi-phase matching is available on such a platform, which gives rise to efficient pair generation. $\chi^{(2)}$ materials have these advantages at the price of larger footprint, higher costs, and reduced fabrication volumes with respect to CMOS technology.¹⁹

A similar choice is open on the actual geometry of the source. Comparing microring resonator and waveguide based heralded single photon sources, the first excel for brightness and do not require spectral filtering, while the second exhibit superior fabrication tolerance and do not need active tuning by means of thermal shifters. In fact, microring resonators have the drawback to be highly fabrication sensitive, due to the coupling gap. The gap, typically around a few hundred nanometers, critically affects the resonating properties of the cavity. Therefore, large scale quantum circuits based on arrays of such microring sources are difficult to scale up, due to the low reliability of the fabrication. To mitigate such an effect, active tuning of the resonance is performed by thermal tuning. Such an operation further increases the already complicated and dense photonic circuit. On the contrary, waveguide based sources have demonstrated higher purities, higher quantum interference visibilities, fabrication tolerance, and reliability. Moreover, their limited brightness can be mitigated by means of multiplexing. Because of this, spiral waveguides, especially with the recent demonstration of the unfiltered dual-pump intermodal source,⁹⁵ are nowadays the best candidates to become the standard source in future integrated quantum devices, being closer to match the fundamental requirements for efficient chip scale quantum operations.¹⁵⁰

Very recently, quantum photonics moved to the mid-infrared (MIR) because of quantum sensing applications and satellite quantum communication.^{151,152} Photon-pair generation and quantum interference experiments have been reported both in silicon waveguides¹⁵¹ and LN crystals.¹⁵² Demonstration of a heralded single photon source integrated in a silicon chip by using the intermodal FWM scheme⁶¹ has been reported.^{153,154} Another avenue to on-chip MIR heralded single photon sources is the use of the induced second order nonlinearity in silicon by lateral p-i-n junctions.^{58,111} However, no reports of SPDC in silicon have been reported yet.

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DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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